

FFAR Final Progress Report

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1 General Information

- 1.1** Please list the geographic location(s) – city, state, congressional district - where the work was conducted.

California: Counties of Monterey, Santa Barbara, San Luis Obispo, Ventura and Imperial.

Arizona: County of Yuma.

- 1.2** How many new jobs were created because of this grant funding?

3 new jobs were created.

- 1.3** How many jobs were maintained because of this grant funding?

3 jobs were maintained.

- 1.4** Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain. N/A – not a non-profit.

- 1.5** Has your organization undergone a recent name change? If so, please provide the new name of your organization. No.

2 Scientific Report

2.1 Public Abstract

Vegetable crop growers lack cost-effective, accurate and easy-to-use irrigation management tools to improve irrigation efficiencies. This project will address the gaps needed to provide such tools to growers and assure them sufficiently accurate irrigation management to maintain crop yields and quality while substantially reducing consumption and runoff. A common method to estimate crop water requirements is to estimate evapotranspiration (ET) for each crop in each field using satellite imagery. However, the spatial and temporal resolutions of satellite imagery limit the accuracy of near real time ET estimates for specialty crops. This project is addressing the effectiveness of supplementing satellite imagery with higher resolution imagery from aerial platforms (aircraft and UAVs) to improve the accuracy of ET estimates for irrigation scheduling. The project also aims to bridge the gap between publicly developed and supported, scientifically based capabilities such as CropManage and NASA's irrigation support tools, by addressing barriers to them being available for integration by commercial technology providers. The project will provide tools and data to growers and the research community, which will lead directly to reduced water consumption and improved water quality while also supporting reporting and compliance with new regulations governing

groundwater pumping and runoff.

2.2 How can the key findings of the project be used by the agriculture community?

The project demonstrated how new advances in UAV technology and airborne sensing can directly impact the quality and timeliness of ET measurements to expand the use of operational irrigation scheduling in the specialty crop industry. By introducing a new, high resolution imagery collection paradigm to support ET models, the available amount of data could grow quickly, which should further advance ET research through data dissemination.

The project also demonstrates how irrigation scheduling capabilities for the specialty crop industry, like CropManage, can be made more user friendly and practical so that industry adoption will increase. This would have a major impact on irrigation efficiency in the western U.S., directly reducing water consumption and improving water quality.

Overall project success will be measured by the degree to which research findings from the scientific community can be transferred to commercial specialty crop producers. The project involved collaboration between growers, grower advocates, academia, and the technology industry in an effort to maximize the chances of adoption by the specialty crop industry. To the extent that this goal is achieved, the project should serve as a precedent for similar, future work associated not only with irrigation practices but also with other water-related issues in agriculture.

2.3 Research Outcomes/Impact

2.3.1 Potential Outcomes

Increased adoption of precision irrigation practices by the vegetable grower industry could reduce water consumption in California and Arizona by these industry sectors and in turn improve water quality through reductions in chemical runoff into groundwater and surface water.

2.3.2 Intermediate Outcomes

During Year 1 of this project, we identified factors causing discrepancies in crop canopy cover measurements that result in uncertainties in evapotranspiration (ET) measurements. These in turn lead to uncertainties in irrigation recommendations, which lead to reluctance on the part of commercial vegetable growers to use irrigation scheduling tools for improved water management.

2.3.3 End Outcomes

The project identified practical obstacles that have limited the adoption of more precise, technology-based irrigation practices in the vegetable production industry. The obstacles were characterized through feedback obtained from industry stakeholders from academia and the commercial sector. The project's field studies characterized the degrees to which uncertainty in fraction vegetation cover measurements and associated evapotranspiration models can impact improved

irrigation practices. It then identified the principal factors causing over irrigation in the production of vegetable crops in the western U.S. It identified and discussed the implications of environmental and regulatory drivers of new irrigation practices throughout the Western U.S. The publicly-available CropManage irrigation scheduling software was expanded to connect to third-party software systems to reduce the requirement of growers entering their data into multiple software systems, which has been a major obstacle to adoption. Lastly, it developed a large set of fractional vegetation cover measurements derived from UAV imagery to be shared with the research community.

2.4 What were the goals/specific aims of the project?

The project schedule was from February 2020 to February 2022, with a no-cost extension granted until August 2022. This duration covers two vegetable crop growing seasons in California (approx. February to November) and two in Arizona (approx. November to March). Hence during the first year of the project, we were able to collect field data during the first project season in California, the second half of the 2019-2020 Arizona season, and the second Arizona season. The overall focus of Year 1 was to establish a technical baseline for making irrigation scheduling improvements and to focus on industry outreach. During Year 2, we focused on additional data collection and usability of software tools to assess means to increase the use of precision irrigation capabilities in the vegetable industry.

Goal 1: Ensure availability of quality data for analysis of ET

Objectives:

- Schedule and acquire UAV and ground-based data in CA.
- Schedule and acquire UAV and ground-based data in AZ.

The objectives associated with this goal involved scheduling the timing of data collection at the field locations determined by customer planting schedules, with the UCCE (CropManage) team advising partner growers on the installation of irrigation flow meters and conducting ET and soil moisture ground truth data experiments using sensors in the project's target fields in California.

In Arizona, the plan included staff from the Yuma Center of Excellence for Desert Agriculture (YCEDA) conducting ET and soil moisture ground truth data experiments using eddy covariance systems and in situ soil moisture sensors.

Goal 2: Analyze ET and evaluate relative accuracy and reliability with different imagery sources.

Objectives:

- Develop data pipeline software to ingest and analyze multiple data sources
- Perform a comparison study of the derived ET measurements.

This goal is meant to be met by comparing the fractional cover derived from satellites to that from higher resolution imagery (from UAVs) to assess the relative accuracy of and effects on ET models. This includes determining how consistent the measurements are at a typical field level (4-5 ha), assessing average effects of non-uniform stand densities on ET estimates using high resolution imagery and incorporating growth model effects (temperature and radiation) to and evaluate their relative effects on near-real-time ET estimates. The objectives associated with this goal are intended to help determine minimal spatial and temporal Fc resolutions required to derive dependable, real-time ET measurements for the targeted crops.

Goal 3: Assess improvements to irrigation scheduling with improved ET

Objectives:

- Add CropManage API features to incorporate improved ET – basic functionality
- Perform CropManage scheduling comparisons with original and improved ET models

These objectives are intended to connect CropManage to third-party data, integrating the “best” Fc values into ET-based water budgeting for the project’s target crops.

Goal 4: Promote and manage stakeholder engagement

Objectives:

- Involve at least 2 grower-partners in ongoing R&D during data collection in their fields (current partner, Bonipak Produce, plus at least one additional grower recruited after the start of the project by the UCCE CropManage team, GeoVisual or WG).
- Organize a project stakeholder advisory committee and seek inputs and feedback from its members as early as possible

These objectives are intended to provide inputs on industry current irrigation practices, viewpoints on what is needed to improve them, and to test and provide feedback on the usefulness of tools developed under this project.

A distributed committee of project stakeholders from the grower and academic communities is intended to advise on the project, with stakeholder engagement sought as early as possible, starting with a project kickoff meeting and continuing with regular meetings and activities.

Goal 5: Stakeholder outreach and feedback events

Objectives:

- Hold a project kickoff meeting
- Hold 1-2 events in California
- Hold 1-2 events in Arizona

Goal 6: Summarize findings from Year 1 and plan for Year 2

Objectives:

- Prepare Year 1 Annual Report
- Hold a Project Team Annual Meeting

2.5 Have any of the major goals/specific aims or milestones for the project changed since the award?

The project's proposed timeline included achieving an initial irrigation scheduling system within the first year that included an API for integration of the capabilities into third party, commercial software solutions. Given the delayed start to trials with CropManage, the team focused the Year 1 effort on the field trials and related data analysis and did not undergo the API extension work until Year 2, when it was completed.

During Year 1, we planned to target lettuce, romaine, and brassicas. Instead, the fields targeted for study with CropManage during Year 1 were romaine and celery. Given a separate celery trial performed by the CropManage team, independent of this project, that was actually advantageous, as described below.

Due to Covid, the Western Growers Center for Innovation and Technology (WGCIT) did not hold any in-person Grower Trial Network (GTN) events as they had planned. GeoVisual discussed with WGCIT the feasibility of holding a virtual event instead, and it was mutually agreed that a virtual GTN event should be held for Colorado's agriculture industry. WGCIT requested Colorado as its first virtual event since it had not yet worked on a GTN activity there but had discussed doing so with the Colorado Fruit and Vegetable Growers Association (CFVGA).

While this project does not include data acquisition and analysis activities for Colorado, the Colorado Fruit and Vegetable Growers Association (CFVGA) is a close collaborating partner with Western Growers, and Colorado State University (CSU) hosts the FFAR-supported [Irrigation Innovation Consortium](#). While Colorado specialty crop acreage is much smaller than California, it is a key state for the industry from a geographic standpoint. Several major vegetable growers based in Salinas also have operations in Colorado, so advances this project brings to California and Arizona would also serve the Colorado market. Also, Colorado has unique water challenges that would be well-served by improved irrigation scheduling, such as potato farming in the San Luis Valley. Given these synergies,

it was concluded that holding the event for Colorado would accomplish two goals: 1) test a virtual format for the GTN on a smaller audience, and 2) initiate a Colorado connection to the western U.S. irrigation work, to be expanded during Year 2.

Bonipak Produce ceased being a grower-partner for this project in September 2020. As such, we will be seeking an additional commercial partner for Year 2 of the project. During Year 1, however, GeoVisual did acquire imagery of sufficient crop fields for this project to meet their cost-share obligation and to provide abundant imagery for analysis, particularly in Yuma, AZ as discussed in Section 1.6.1.6, below.

2.6 What was accomplished under the goals/specific aims or milestones for the project?

2.6.1 Goal 1 – Ensuring availability of quality data for analysis of ET

2.6.1.1 Summary of Planned Objectives and Activities for this Goal

The principal objective for this goal was to acquire UAV and ground-based data in California and Arizona. Our plan involved coordinating UAV imagery acquisition with both customer planting schedules and the University of California’s Agriculture and Natural Resources (UCANR) CropManage team’s irrigation trial activities with the same customers. In Arizona, the plan included using staff from the Yuma Center of Excellence for Desert Agriculture (YCEDA) to conduct ground truth data collection.

2.6.1.2 Summary of Actual Objectives Achieved and Activities Conducted for this Goal

We originally planned to expand the variety of crops we studied during Year 2 of the project, but instead we continued our focus on leafy green crops during Year 2. This afforded us the opportunity to obtain the largest amount of data with broad geographic coverage, which was a request from FFAR during the project’s Year 1 Annual Report review. A focus on leafy green crops also was most relevant for the project to provide a regional assessment of the irrigation issues affecting the same, or similar, crops, with leafy green crops grown principally in the Salinas Valley and Central Valley of California for about $\frac{2}{3}$ of the year and then in Yuma, Arizona for the remaining $\frac{1}{3}$, with some limited exceptions. Lastly, we were also constrained by which crops were of interest to our grower partners and which growers were open to working with the UCANR’s CropManage system. With Nature’s Reward Farms remaining the most committed customer-partner, we gained the most insights from their explorations of new irrigation practices to improve efficiency.

2.6.1.3 Aerial Imagery Acquired for Evaluation of Fractional Vegetation Cover

During Year 2, we coordinated with several customer partners to obtain aerial imagery and compute fractional vegetation cover (Fc). Figures 1 through 3 indicate the geographic locations in California and Arizona of the fields over which GeoVisual acquired multispectral images (4 channels: red, green, blue and near infrared) using a UAV (“drone”) platform to derive Fc. The drone was flown to acquire single image frames with a ground sampling distance (GSD) of 1.0 cm and sufficient overlap and side lap between images to support photogrammetric processing. A single frame covered about 80 meters cross-track and 50 meters along track, for a total area of about 1.25 acres.

The complete dataset includes 325 independent imagery collections over iceberg lettuce fields from October 1, 2020, through November 18, 2022. Most fields were imaged once, with some imaged twice during the crop growth cycle. This dataset will be made available publicly as a deliverable product from this project. In Section 2.6.2.3, we demonstrate uses of the dataset to support other studies for regional evapotranspiration measurements.

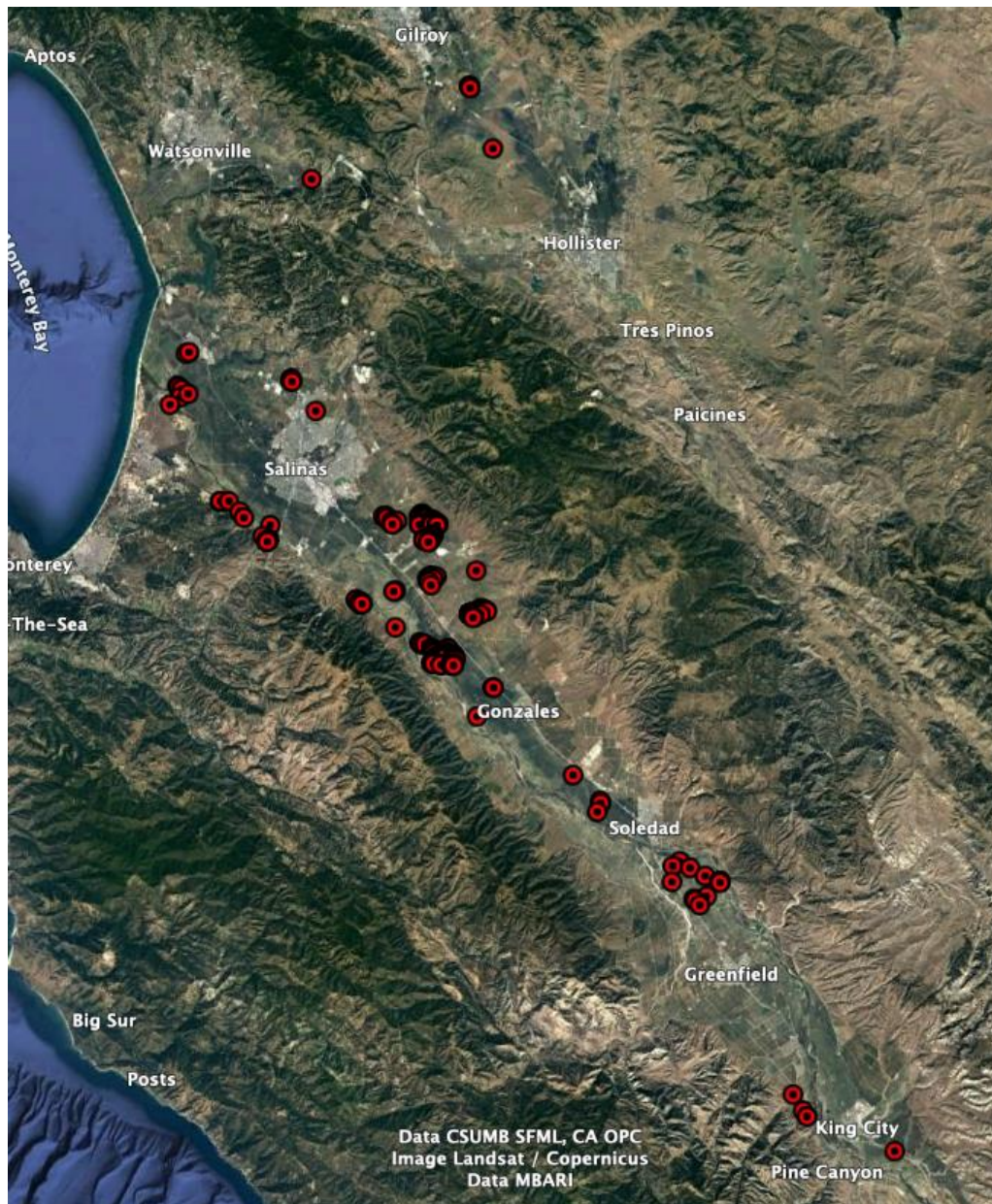


Figure 1. Geographic locations (red markers) of fields throughout the Salinas Valley region over which GeoVisual flew UAV imagery to derive F_c . The complete dataset includes 325 independent imagery collections over iceberg lettuce fields from October 1, 2020, through November 18, 2022.



Figure 2. Geographic locations (red markers) of fields throughout the Yuma Valley region and next to the Salton Sea over which GeoVisual flew UAV imagery to derive fractional vegetation cover.



Figure 3. Zoomed in view of the geographic locations (red markers) of fields throughout the Yuma Valley region over which GeoVisual flew UAV imagery to derive fractional vegetation cover.

For each field, we derived the F_c by first, calculating the area of ground each plant in the field covers, summing all the areas, and then dividing by the total field area to get the average F_c value of the field. With this method of computing F_c , we also derived the F_c statistics given the statistics of the ground area cover of each plant. In the following section, we discuss using the 95th percentile F_c - the value for which only 5% of the plants in the field were larger - to ensure ET estimations are based on the densest vegetation in a field so as to not underestimate irrigation requirements.

Figure 4 shows the distribution of 95th percentile Fc values for all 325 fields analyzed. Most of the fields were imaged with Fc values between about 0.2 and 0.4, which corresponds roughly to the mid-point of the growth cycle.

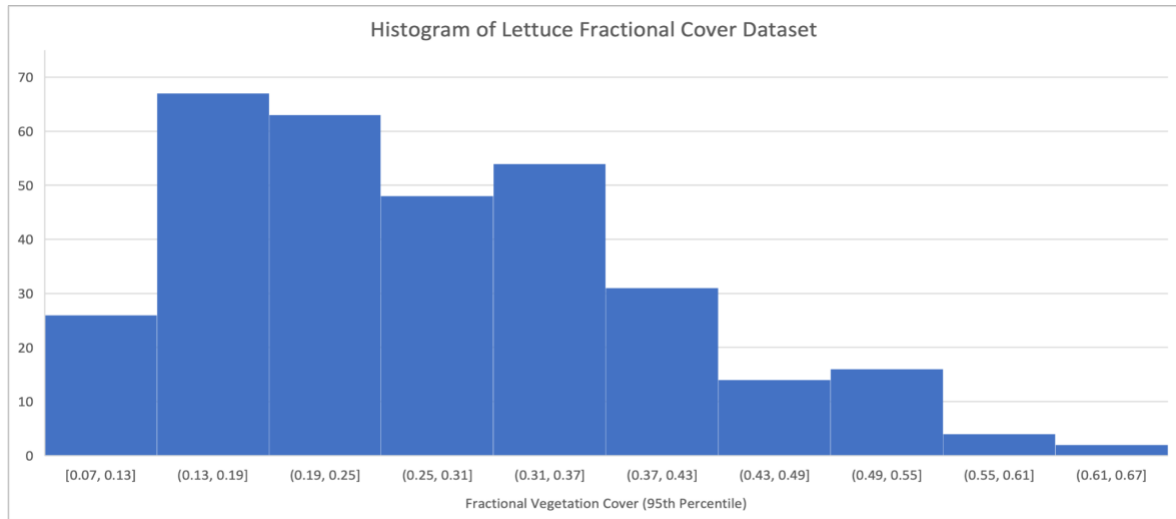


Figure 4. Histogram of Fc values for the 325 lettuce fields imaged with GeoVisual's UAV in California and Arizona. Most of the fields were imaged with Fc values between about 0.2 and 0.4, which corresponds roughly to the mid-point of the growth cycle.

2.6.1.4 In-Field Measurements of Applied Irrigation

Year 2 of the project involved continued work with Nature's Reward and the UCANR to investigate means to gain adoption of irrigation scheduling. With funding from the California Department of Food and Agriculture's (CDFA) program, the State Water and Efficiency Enhancement Program (SWEET) grant, Nature's Reward worked with extension staff to install systems of flow meters and pressure regulators in such a way as to incentivize irrigators to more precisely apply irrigation water and fertilizer. A common configuration in vegetable fields due to the quick crop turnaround, is a small trailer with a water filter that connects to the main line, with the trailer allowing portability between fields. The advancement made during Year 2 of this project was to configure a trailer with a flow meter and pressure regulator in addition to a filter, for a trailer configuration as shown in the example in Figure 5. The pressure regulator is required to ensure proper application of the amount of water prescribed by an ET-based irrigation recommendation. The flow meter data can be collected with a data logger and accessed remotely to monitor the irrigation system. But even without constant monitoring, flow meters on a few fields can be used to get "actuals", with these

used as estimates for the flow on other fields with similar irrigation systems.

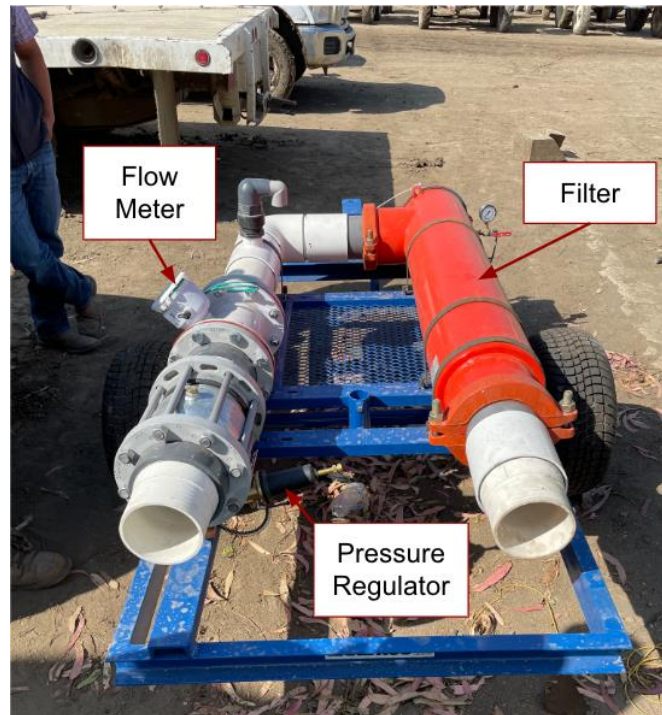


Figure 5. Irrigation trailer configured with a water filter, pressure regulator, and flow meter to be used for precision drip irrigation. The flow meter data can be collected with a data logger and accessed remotely to monitor the irrigation system. Funding to build the complete irrigation trailer was provided to Nature's Reward Farms by a CDFA SWEEP grant. Adoption and use of the trailers were a subject of this project in Year 2.

2.6.2 Goal 2 – Analysis of ET and Relative Accuracy and Reliability with Different Imagery

2.6.2.1 Summary of Planned Objectives and Activities for this Goal

The principal objectives under this goal are to compare canopy F_c estimates from different imagery sources and determine what impacts canopy cover differences have on the resulting estimates of ET and recommended irrigation schedules. The overall intent is to help determine the spatial and temporal canopy cover resolutions required to derive dependable, real-time ET measurements for the targeted crops, as part of a practical methodology that growers would adopt.

2.6.2.2 Summary of Actual Objectives Achieved and Activities Conducted for this Goal

During Year 1 of the project, we completed a comparison of canopy cover estimates from imagery from satellites, a UAV and in-field cameras for lettuce and celery. We also examined the differences in the measurements from different

sources and made some preliminary conclusions about the effects of field variability and the spatial sampling extents of the difference sources. To study the impacts of canopy cover errors on ET estimates and irrigation schedules, we modeled ET using the established formula used by CropManage and simulated irrigation schedules with canopy errors taken into account. During Year 2, we addressed the intra-field variability issues identified during Year 1 and then focused on issues related to adoption of satellite-based ET estimates for broad area ET tracking.

During Year 1 of the project, as reported extensively in the Annual Report, we found that most fields are highly non-uniform, with a range of Fc values across the field. Furthermore, we encountered a discrepancy between the Fc values reported by CropManage and those GeoVisual derived. When the UCANR CropManage team performs field trials, they routinely also acquire a set of photos from a multispectral camera held on a pole about 5 meters above the target field. These are used to evaluate the fractional vegetation cover at the time of the photos and calibrate the modeled growth curve. Generally, the average Fc values GeoVisual derived from the UAV imagery were smaller than the in-field estimates obtained by the CropManage field teams. CropManage measurements typically were collected in denser areas of each field since it is important to derive associated ET estimates for the largest plants in a field, to ensure the field is not under-irrigated. We confirmed this by comparing the CropManage in-field measurements with UAV-derived values for the 95th percentile Fc: the value for which only 5% of the plants in the field were larger.

These 95th percentile Fc values matched the CropManage in-field values closely, as demonstrated by the Fc plots in Figures 4 and 5. In contrast, the average UAV-derived Fc field values shown in Figures 6 and 7 for the same fields are significantly lower (~ 10-15%) than the in-field measurement values ("User Data").

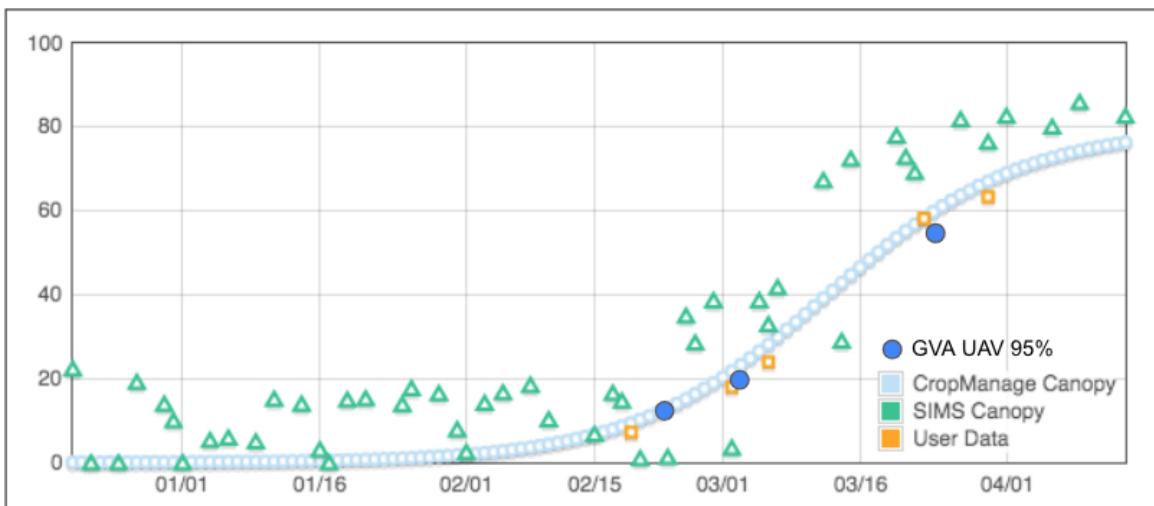


Figure 6. CropManage plots of canopy cover model and measurements for lettuce field Callaghan 41-3. The vertical axis is the percentage of the field covered in

vegetation versus bare soil (fractional cover, F_c). The dates on the horizontal axis are for 2021. The logistic function model is shown in light blue, estimates from NASA's Satellite Irrigation Management System (SIMS) are shown as green triangles, in-field measurements using a multispectral camera held above the field are shown as orange squares ("User Data"), and 95th percentile F_c values from GeoVisual's "ground area cover" (GAC) processing are shown as blue circles.

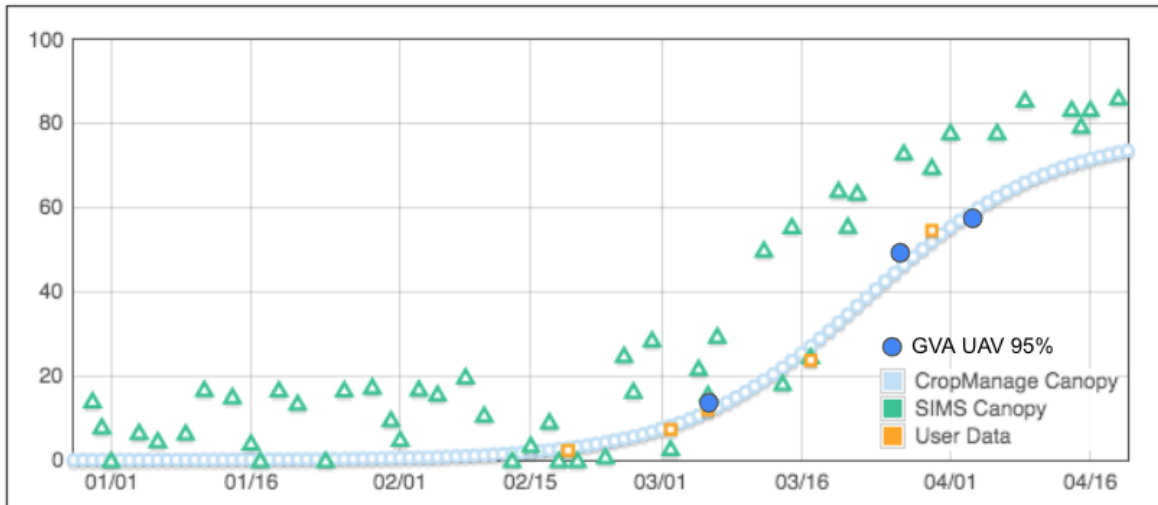


Figure 7. CropManage plots of canopy cover model and measurements for lettuce field Callaghan 41-4. The vertical axis is F_c percentage, and the model and measurement labels are the same as in Figure 4.

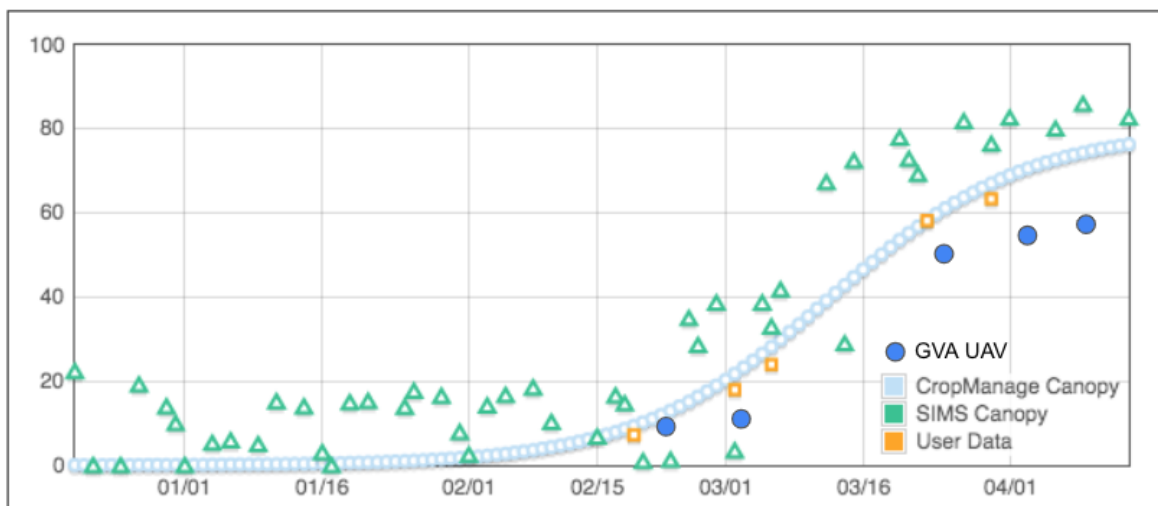


Figure 8. CropManage plots of canopy cover model and measurements for lettuce field Callaghan 41-3. The vertical axis is F_c percentage, and the model and measurement labels are the same as in Figure 4 except the GVA F_c values are the

average Fc values and not the 95th percentile. Notice the average values are consistently lower than the in-field ("User Data") values.

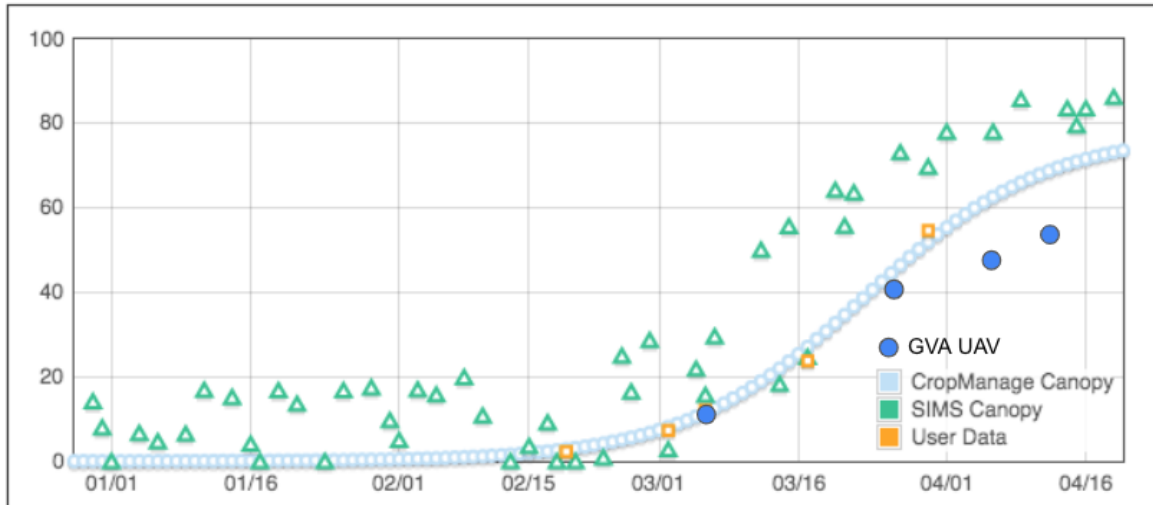


Figure 9. CropManage plots of canopy cover model and measurements for lettuce field Callaghan 41-4. The vertical axis is Fc percentage, and the model and measurement labels are the same as in Figure 4 except the GVA Fc values are the average Fc values and not the 95th percentile. Notice the average values are consistently lower than the in-field ("User Data") values.

Note that the Fc values derived from satellite imagery in NASA's SIMS system are consistently high in Figures 6 through 9. The SIMS canopy cover values are derived from established relationships between Landsat NDVI values and Fc values obtained from field photo measurements (Trout, Johnson and Gartung 2008), (Johnson and Trout 2012). The satellite derived NDVI values are known to generally overestimate the canopy cover for low cover values but are considered accurate for higher values (M. Cahn, personal communication). One possible source of bias is that wet (darker) soil elevates NDVI values (Johnson and Trout 2012). As reported by Michael Cahn of CropManage (personal communication), an error was found in the API code that provides the SIMS values to CropManage, and this error was present in Q1 of 2021 when the data was pulled from SIMS for Figures 4 through 7. Later access to SIMS is said to provide more accurate values. Our set of UAV-derived Fc values will be shared with the NASA SIMS team as a means for them to validate their service.

2.6.2.3 UAV and Satellite-derived Fractional Cover Analysis

During Year 2, we collaborated with the University of Arizona (U of A) and USDA/ARS on a comparison of UAV-derived Fc and satellite-derived Fc from vegetation index modeling. The U of A project, entitled Quantitative Assessment of Water & Salt Balance for Cropping Systems in Lower Colorado River Irrigation Districts (henceforth referred to as the Arizona Quantitative Assessment project),

is an ongoing collaborative research project coordinated by the U of A's Yuma Center of Excellence for Desert Agriculture (YCEDA) and involving the USDA/ARS office in Maricopa, Arizona. Its focus has been on measuring the precise amount of water today's crops use and studying the impacts of irrigation, crop rotation and fallowing on soil salinity.

The basis for that project was a case study in water use efficiency in the region, with an objective of updating the models of water use for economically significant crops in the Yuma Valley and tracking salinity impacts resulting from crop irrigation there. For the water use study, updated crop coefficients were developed from satellite imagery, to relate reference ET values to crop-specific ET values for 17 crops grown in the region. The driver for updating the consumptive use of these crops originally was interstate concerns over use of Colorado River water.

The UAV imagery collected under this FFAR project was used to compare Fc values derived from the satellite imagery study. The USDA/ARS team lead on the project, Dr. Andrew French, did some preliminary comparisons using the UAV imagery collected during Year 1 and will be incorporating the much larger Year 2 Fc data set in a future publication. Results of the preliminary comparison are shown in Figure 10. Fc values are shown on the y-axis, derived from both GeoVisual's UAV imagery ("GV" red data) and YCEDA-acquired UAV imagery ("YCEDA" light blue data). The NDVI values on the x-axis were derived from Sentinel-2 satellite imagery collected on the same day and at the same locations. Note that the 10 m satellite pixel size effectively averages large sections of each field, so for these analyses the GVA average Fc values may be a better comparison than the 95th percentile values.

Also, of note in Figure 10 is that the GVA UAV-derived Fc values are limited to below about 0.6 given the timing of our imagery collections, whereas the YCEDA data was acquired for higher Fc values. Dr. French pointed out there are operational challenges to obtaining a time series of images that span a broad range of Fc values for a single field, so combining a larger set of different field measurements can be useful.

Figure 10 also indicates that although there is a reasonable linear fit between NDVI and Fc for the given range of Fc values, with an R² value of 0.92, it demonstrates that a smaller slope would fit the data better for smaller Fc values. Since NASA SIMS uses a linear relationship between NDVI and Fc with a slope of 1.26, this may at least partially explain why SIMS overestimates actual Fc at low values. As summarized in this project's Annual Report, SIMS Fc values typically are 10-20% higher than our UAV estimates.

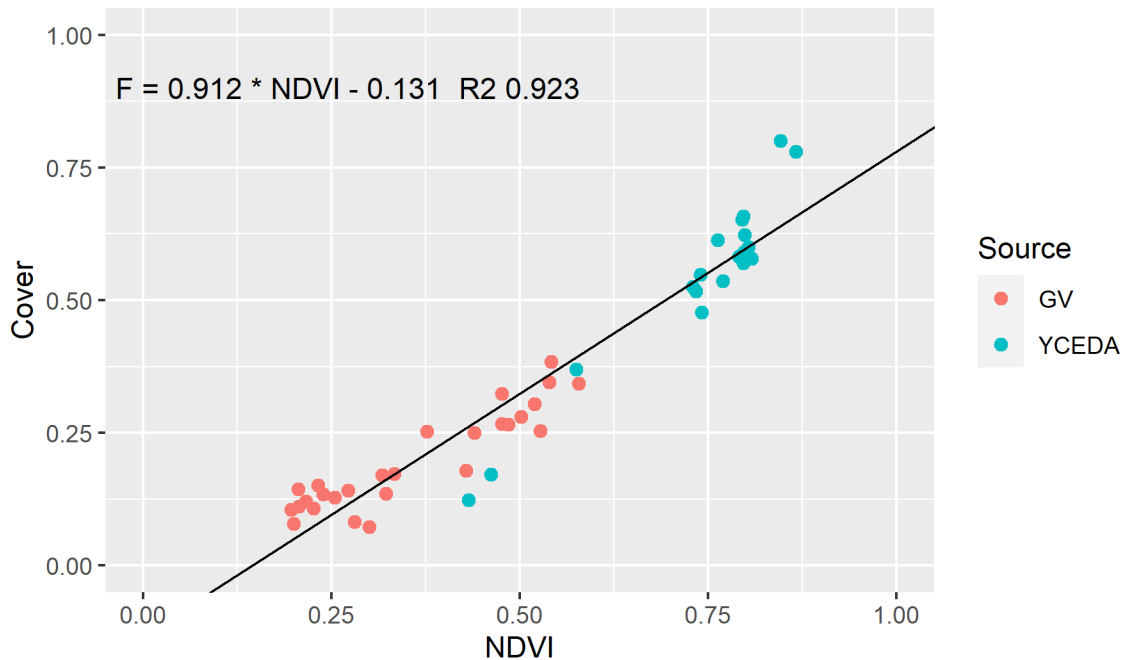


Figure 10. Fit of UAV imagery-derived fractional vegetation cover of iceberg lettuce (GVA results in red and YCEDA data in blue) versus normalized difference vegetation index (NDVI) values derived from Sentinel-2 satellite imagery collected on the same day and at the same locations. This work was led by the USDA/ARS team as part of the Arizona Quantitative Assessment project. USDA/ARS plans to continue this study using the full set of GVA fractional cover data for the locations illustrated in Figures 1 through 3.

Another source of error in using satellite imagery to estimate ET is how annual crops like vegetables must be treated. Because of the lack of real-time knowledge about crop type in any particular field, SIMS makes use of data from the Cropland Data Layer to identify fields that are assumed to be cultivated with annual crops (no more specific information is available) and makes use of a generalized Kc-Fc relationship based on a best-fit for several crops (from Allan and Pereira, 2009), as shown in Figure 11 (from Melton et al., 2012). This generalization is acknowledged by the authors to introduce a crop coefficient estimation uncertainty of up to 14%. In (Pereira, et al. 2020), comparisons of the SIM model results for ET based on its crop coefficient models to ground-based ET datasets (e.g. from eddy covariance stations) indicate a mean absolute error of 15% across all crops. Using a soil water balance model to correct for soil evaporation and crop water stress reduced the seasonal mean average error to less than 10%.

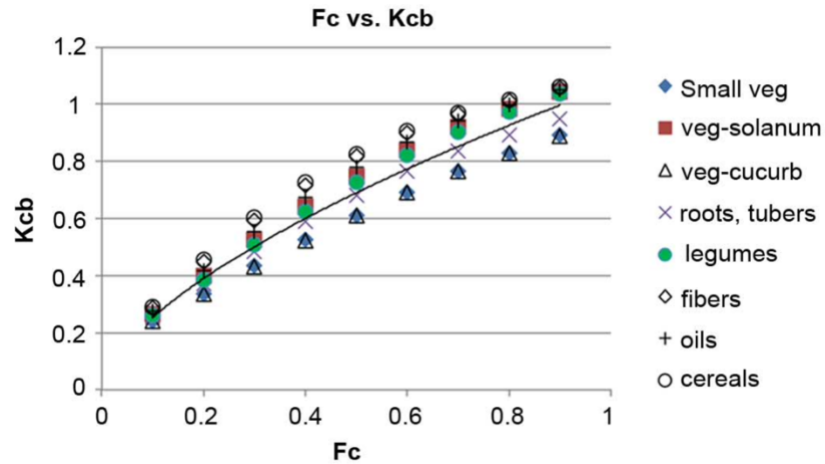


Figure 11. In SIMS, a best-fit to crop-specific relationships is used to convert fractional cover to a crop coefficient for annual crops, for which the actual crop type is not known. This introduces up to 14% error as acknowledged by the authors (Melton, et al.).

In (Pereira, et al. 2020), it is mentioned that SIMS allows the user to specify crop type information as an input to the model. While accessing SIMS through its API for testing was outside the scope of this project, this capability would reduce some of the error caused by using the best-fit relationship indicated in Figure 11.

Notwithstanding this capability for SIMS, reliable, frequently updated data on ET for vegetable crops is not, at the date of this report, available on a widespread basis. The OpenET initiative shows promise as a future data source, particularly when API access is available for connectivity to irrigation scheduling and management systems. However, as of the date of this report it does not properly handle crop type for annual crops, at least through the browser interface. An example is provided in Figure 12, where a field boundary is selected in the OpenET browser interface, and the crop type is displayed erroneously as “Grass Hay” in the pop out box with the monthly ET data for 2022. This same field is one for which UAV imagery was acquired by GeoVisual in October 2022 and was known to contain a lettuce crop. The discrepancy can be traced to OpenET’s reliance on the Cropland Data Layer, which is known to have a high error rate for annual crops and particularly for vegetable crops that typically contain at least 2 plantings, often of different crops, each year.

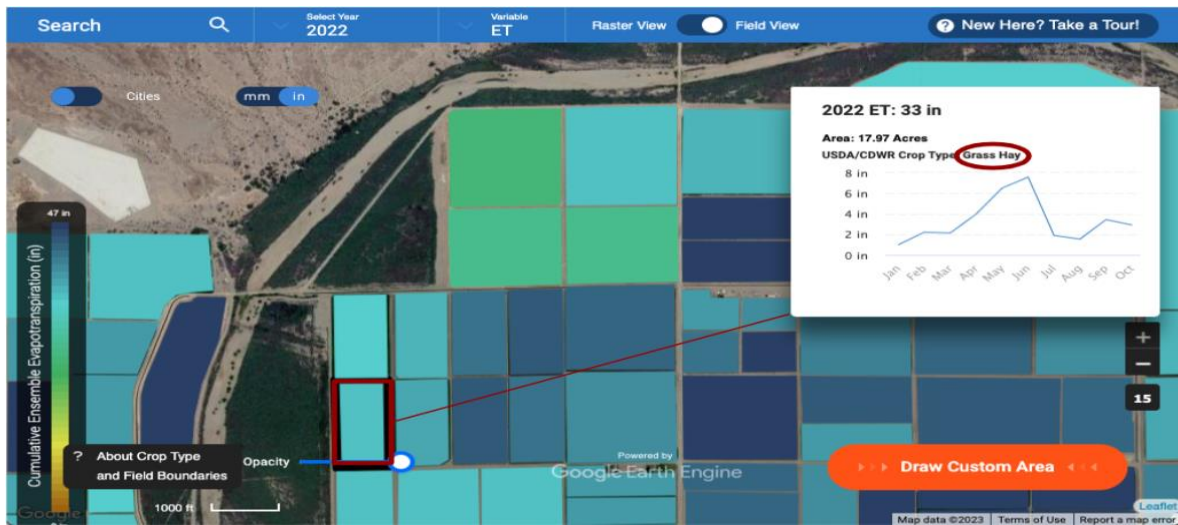


Figure 12. OpenET web interface access to 2022 monthly ET data for a field near Yuma, AZ, with the crop type given as Grass Hay. This field was known to contain lettuce in October, 2022 (see Figure 13).

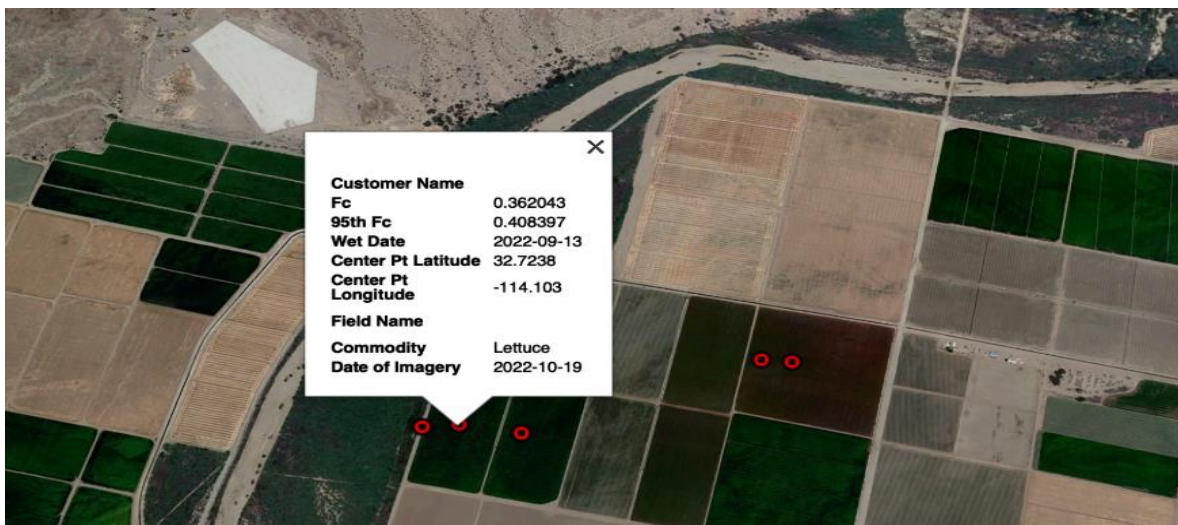


Figure 13. Google Earth view of the same field as shown in Figure 12, with the metadata from GeoVisual's UAV imagery collection shown. This field contained lettuce in October 2022.

2.6.2.4 Conclusions from Canopy Cover Studies and Operational Implications

When using canopy cover to estimate irrigation requirements via ET, the balance to be found is between 1) ensuring sufficient irrigation for the most vigorous

sections of a field and 2) avoiding water waste, degraded water quality from fertilizer runoff, or crop degradation from disease.

All of the in-field, pole-camera measurements were performed for irrigation trials, with CropManage staff collecting and analyzing the images to derive F_c . Our conclusion is that in-field measurements of canopy cover are not sufficiently practical for growers to perform by themselves in production, given the level of specialized training needed and image collection time required and presently an insufficient value proposition to warrant the effort. Therefore, remotely sensed measurements either from a UAV or, ideally from satellites, are favorable, assuming the data can be provided to CropManage or another irrigation scheduling tool to provide accurate and dependable ET estimates. Local weather stations or data from statewide networks, such as CIMIS in California, AZMET in Arizona or CoAgMET in Colorado, are required to provide reference evapotranspiration (ET_0) values.

We summarize our analysis above as follows:

- Remotely sensed imagery can overcome the time and labor barrier of collecting in-field imagery estimates of F_c , provided there is sufficient added value to warrant the remotely sensed imagery (in the case of commercial UAV imagery), or the remotely sensed imagery is freely available (in the case of non-commercial satellite imagery) and is integrated into an irrigation scheduling system.
- Using 95th percentile measurements of canopy cover derived from UAV-based or other high-resolution imagery provides a conservative estimate of a field's canopy cover that will ensure ET estimates account for the transpiration demands of the larger plants in the field.
- The uncertainty in satellite-derived estimates of canopy cover for vegetable crops is in the range of 10%-20%, depending on the crop type and with more uncertainty at lower F_c values. The sources of error include assumed linear fits between vegetation indices and F_c values, a bias from wet (darker) soil that elevates NDVI values, and the averaging of crop coefficient values from a set of annual crops since real-time, crop-specific information is not available through the Cropland Data Layer or other sources for annual crops.
- These conclusions suggest research gaps to be investigated further: improved F_c cover estimates from satellite-derived vegetation indices like NDVI for low F_c , and improved ET models for low F_c value fields, which are mostly bare soil.
- While work in these areas is ongoing, there still is no reliable, widespread source for vegetable crops to be used operationally.

2.6.2.5 Impacts of Canopy Cover and ET Model Uncertainty on Irrigation Scheduling

In this project's Year 1 report, the ET model for CropManage (Johnson, Cahn, et al. 2016) was used to evaluate the impacts on irrigation scheduling of F_c and crop coefficient estimation errors. As discussed in the Year 1 report, the standard crop coefficient approach to modeling ET was used, where the crop coefficient K_c is a linear scale factor relating crop-specific ET (ET_c) to reference ET (ET_0):

$$ET_c = ET_0 \times K_c \quad (1)$$

ET_0 is derived from one of a number of standard energy balance equations and daily weather data (Temesgen, et al. 2005). In CropManage, the crop coefficient K_c is separated into a transpiration component T and an evaporation component K_e , and K_c is assigned whichever component is greater over time:

$$K_c = \max [T, K_e] \quad (2)$$

The canopy cover is modeled with a logistic function (as shown in the canopy cover plots above), and T is modeled as a second order polynomial function of canopy cover (Equation 2 in (Johnson, Cahn, et al. 2016)). For the evaporation component of ET, CropManage distinguishes between drip and sprinkler irrigation, since drip irrigation doesn't wet the entire field surface, and it follows the work of (Gallardo, et al. 1996) in specifying decaying values of K_e over time after irrigation events.

These models were used with weather data from a station in the Salinas Valley to examine the effects of different canopy growth values on the resulting, cumulative ET values over time. We looked at two scenarios with different canopy growth curves and used the cumulative ET values from historical weather to compare the irrigation requirements. If one scenario represents the actual canopy cover growth over time and the second one represents an erroneous estimate of the canopy cover, then the comparative irrigation requirements indicate the impact of canopy cover measurement errors on the resulting irrigation recommendations.

From this model study and our updated findings regarding imagery estimates for ET using UAVs and satellites, we drew the following conclusions regarding the practical implications for irrigation scheduling:

- Until the crop reaches a maturity at which transpiration dominates, differences in fractional cover make little difference since ET is dominated by bare soil evaporation after sprinkler irrigation events. This corresponds to canopy cover values of about 25% for the models examined for lettuce and broccoli.
- Once transpiration dominates, errors in canopy fractional cover cause a similarly sized error in ET estimation and thereby irrigation requirements, reflecting the linear relationship between canopy cover, crop coefficient and ET for larger canopy cover values. Generally, each 10% error in fractional

cover leads to just under a 10% error in the irrigation requirement, depending on the crop and how long it takes for the canopy cover to fill in.

- Given the current ET error levels from satellite sources, as discussed in the previous section as in the range of 10-20%, and our conclusions regarding associated irrigation requirements, vegetable growers could safely provide a buffer to account for them (e.g., applying irrigation amounts with an additional 20% over what is recommended by satellite ET estimates).
- In the following section, we discuss the implications of these findings on perceived priorities for improving irrigation scheduling practices in commercial vegetable production.

2.6.3 Goal 3 – Assessment of Improvements to Irrigation Scheduling with Improved ET

2.6.3.1 Summary of Planned Objectives and Activities for this Goal

The objectives under this goal are intended to integrate the best possible canopy cover values into irrigation scheduling capabilities. They include incorporating improved ET into CropManage and performing scheduling comparisons. However, recommendations for efficiently obtaining canopy cover and comparisons of the effects of improved canopy estimates on irrigation scheduling are independent of CropManage. Therefore, the observations and conclusions we have made under this goal apply to irrigation scheduling in general.

2.6.3.2 Summary of Actual Objectives Achieved and Activities Conducted for this Goal

The objective of integrating the best possible canopy cover values into irrigation scheduling capabilities is not complete at the end of Year 1, since we are still assessing the discrepancies, we observed in canopy cover measurements (discussed in Section 1.6.2 above, under the previous goal). The canopy cover errors and associated comparisons of their impacts on irrigation scheduling that we made and discussed in Section 1.6.2.9 are the framework under which we can now compare additional fractional cover measurements and scheduling impacts, for additional lettuce and celery fields and for additional crops.

The objectives for this goal also included connecting CropManage to third-party data via its API. Based on what we learned under Goal 2, we concluded that the existing capability for including third party data into CropManage is sufficient for current studies. If the underlying canopy cover model is sufficient, only a handful of third-party data points are required to properly calibrate the model.

Currently in CropManage the parameters that adjust the shape of the logistic canopy model are adjusted manually in the user interface. A more efficient and

quantitatively rigorous way to adjust the model would be to fit it algorithmically to available data, including SIMS and any additional third-party data.

2.6.3.3 Addressing Excess Irrigation During Early-Stage Growth

We began this goal by establishing a baseline for improvements to water use in commercial vegetable production today. Through interviews with growers and UCANR, we concluded that much of the over-irrigation in fresh vegetable crops is done in the early periods of growth, after seeding or transplanting, when growers are concerned with ensuring sufficient seedlings or transplants survive to provide the target plant spacing along each row (a good “stand”).

To quantify these reports for the lettuce and celery fields we studied during Year 1, we compared the CropManage irrigation recommendations to actuals for three fields: a lettuce field germinated with sprinkler irrigation (which is most typical), a lettuce field germinated with drip irrigation (not typical), and a celery field germinated with drip. The cumulative irrigation amounts as a percentage of CropManage recommendation for the three fields were 230%, 392% and 151%, respectively. Possibly the grower applied a larger excess amount on the drip germination lettuce field than the sprinkler germination field since drip germination is far less common than sprinkler germination for lettuce.

Irrigations during the first 2 weeks after planting were the ones with the largest excesses over recommendations, consistent with the UCANR observation that grower’s overwater to ensure a good stand “up front”. This practice results in a large portion of the total water applied to a field being applied in the earliest periods. For all three of the test fields, at least 40% of the total applied water was applied in the first 2 weeks after planting, and over 50% was applied within the first 30 days. It’s possible that the largest savings through reductions in over-irrigation are realizable in the first 2 weeks to 30 days after planting.

During these early stages of crop growth, plants are very small, and ET is principally driven by evaporation from bare soil. Therefore, irrigation amounts can be based on evaporation replacement only, with no need for fractional cover measurements from imagery to estimate transpiration. But reducing early overirrigation, principally viewed by irrigators as “insurance” against poor germination or transplant die off, requires more precise water application and monitoring capabilities than are being used today. Therefore, a principal focus for Year 2 of the project was to investigate operational techniques to increase adoption of technologies and practices that will make irrigation more precise and verifiable.

Emphasis within the industry and by UCANR's irrigation teams has been to get pressure regulators fit into commercial irrigation systems, since these are required to effectively use drip irrigation and to track actual amounts of applied water. Another related requirement is to ensure the proper run lengths for drip tape, since if drip tape lengths are too long the water pressure becomes too low for proper irrigation towards the end of the line.

Adoption of pressure regulators by irrigation crews proved non-trivial. A survey of irrigation crews indicated that they fear setting system pressures too high will cause drip tape hoses and connectors to explode or disconnect, so they tend to keep pressures as low as possible. Hence there is the inertia of current and past practices that must be overcome through demonstrations and sufficient experience. It was confirmed by UCANR staff that these attitudes and practices are not isolated to Nature's Reward crews, with whom we worked most closely. It is common in the industry. More often than not, the issue is that the drip tape systems are not configured properly. When connectors are configured properly, they do not have problems. However, because irrigators configure systems in each field and then move on to other things, they are not around to observe when disconnects happen and therefore fear a system will disconnect and the irrigation will not proceed properly. Hence system integrity must be improved; there has been discussion of using "spin lock" connectors to ensure a more fail-safe system.

Flow meters are also required for precise irrigation, but typical vegetable field production today does not include them. In most cases, at least for the companies participating in this project, flow meters only exist on well heads that provide water to many fields. The reason is cost. It would be very expensive to have flow meters on individual fields when a typical mid-sized vegetable operation can have over a hundred drip systems operating at any given time. Therefore, it is not possible today to know how much water flows to each individual field.

The solution devised for Nature's Reward, in collaboration with UCANR, was to configure the filter trailer shown in Figure 5 above. The trailer can be moved between neighboring fields when they are to be irrigated, ensuring a pressure regulator and flow meter during each irrigation. Results during the first season of use indicate that irrigation crews like using the trailers because they save them time, with less irrigation infrastructure to connect and configure for each field. Nature's Reward is taking the approach of providing the limited number of trailers available only to the irrigation crews that use them to properly configure their drip irrigation systems at the correct pressures and with fail-safe fittings. They have thereby created an incentive process that rewards good irrigation practices with time savings. Nature's Reward built 10 irrigation trailers with funding from the

CDFA's SWEEP program in 2022. They plan to seek additional SWEEP funding for additional trailers in 2023.

Incentivizing irrigators to adopt more precision systems as is being done by Nature's Reward will not by itself solve the early-stage over-irrigation problem, but it is the first step towards addressing it. Once irrigators have a system whose performance and dependability they can trust and that provides them knowledge of the precise amount of water applied, they may be more compelled to apply less "insurance" (excess water) to each field. But decisions on how much water to apply, when, and the data on how much water is actually applied all must be tied into the overall operational process of crop irrigation and fertilization. Addressing how to modernize that process was another focal point of this project, which is discussed next.

2.6.3.4 Irrigation & Fertilization Process Improvements to Incorporate New Technologies

To evaluate steps to be taken to modernize irrigation and fertilization, we developed process flow diagrams for all aspects of a mid-sized vegetable production operation related to irrigation and fertilization. The resulting flow diagrams are shown in Figure 14 and Figure 15, for irrigation and fertilization, respectively. During the project, the "Irrigation Recommendation" and "Fertilization Recommendation" boxes correspond to UCANR's CropManage system, but these diagrams are intended to be general, for different software systems, growers and crops grown. Note that existing data flow pathways are shown as black lines, while new, recommended pathways are shown in purple.

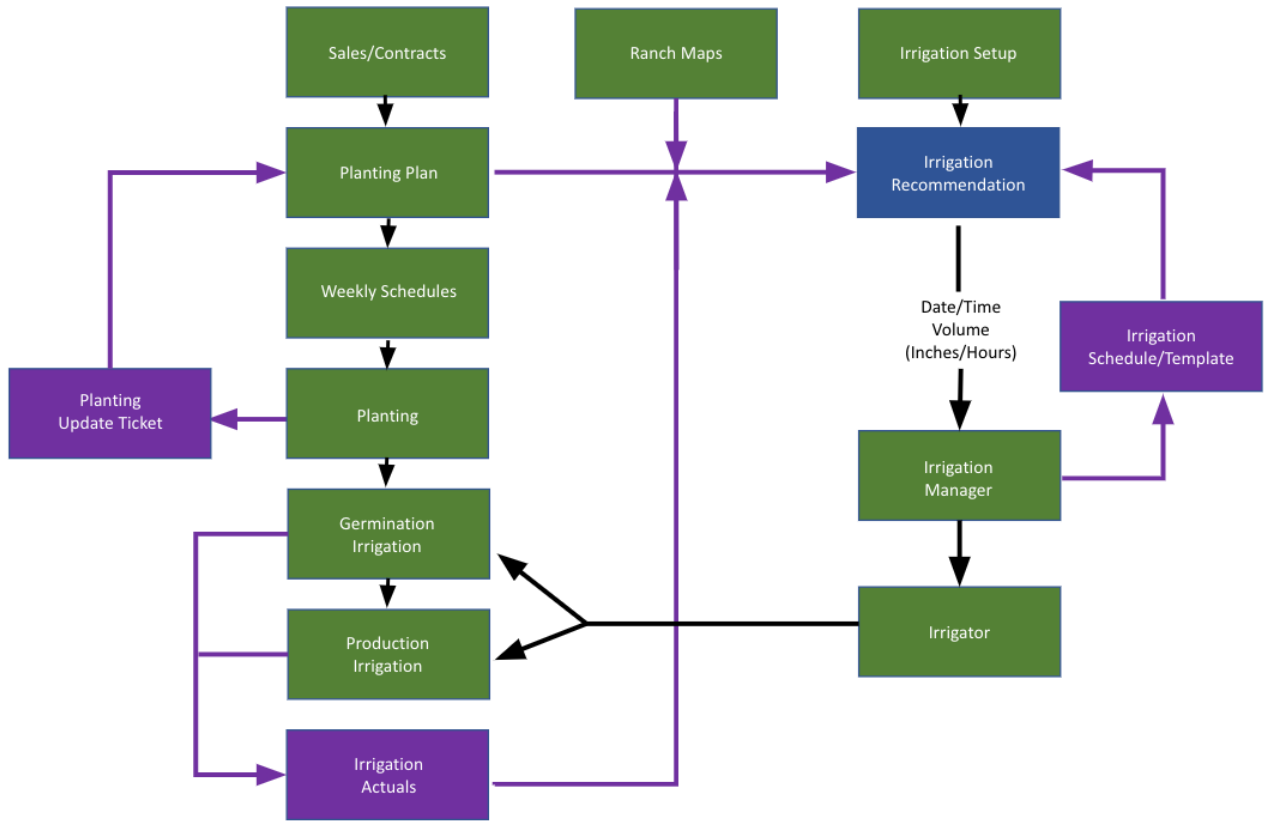


Figure 14. Irrigation process map for commercial vegetable production with additions (in purple) to incorporate precision irrigation recommendations.

Walking through the Irrigation process map from the upper righthand corner in Figure 14, Irrigation Setup refers to identifying the planned infrastructure such as pipes and nozzles and inputting associated data on it that influences the irrigation recommendation into the Irrigation Recommendation system. The planned irrigation schedule also is input to the recommendation system, so that the recommender will determine how much to apply on the scheduled dates, depending on the weather conditions and soil type. The output from the recommendation system is provided in terms of a rate (e.g., inches/hour) or as a total amount (volume) that then can be converted into an irrigation time for a given infrastructure that is known to apply a certain rate.

Information on the fields, including their size and the crop and variety to be planted, also is provided to the irrigation system, since the ET model is dependent on crop type. These come from available ranch maps and planting plans. Note we have added the concept of a “Planting Update Ticket”, since if plans change then the irrigation system should be able to access the changes to account for any changes in planting dates, different crops, etc.

Lastly, the irrigation recommendation system requires an update on the actual amount of water applied at every irrigation event, in order to make the next recommendation and properly account for ET replacement requirements.

Similarly, in Figure 15 for fertilization, the irrigation setup must be provided to the recommendation system, since if it is a drip system then fertigation will be applied, along with the plan for fertilization (e.g., type of fertilizer applied and schedule). If fertigation is not used, then teams that handle equipment (e.g., those scheduling tractors) must be involved. Similar to irrigation scheduling, the recommendation system also requires field information from ranch maps and the planting plan, and actual fertilizer applications should be recorded for regulatory reporting. Typically, soil tests are performed before each fertilization event to help determine the amount to apply. Equivalent to irrigation scheduling, we have added the concept of a “Planting Update Ticket” so that the irrigation system can access any changes to planting dates, different crops, etc.

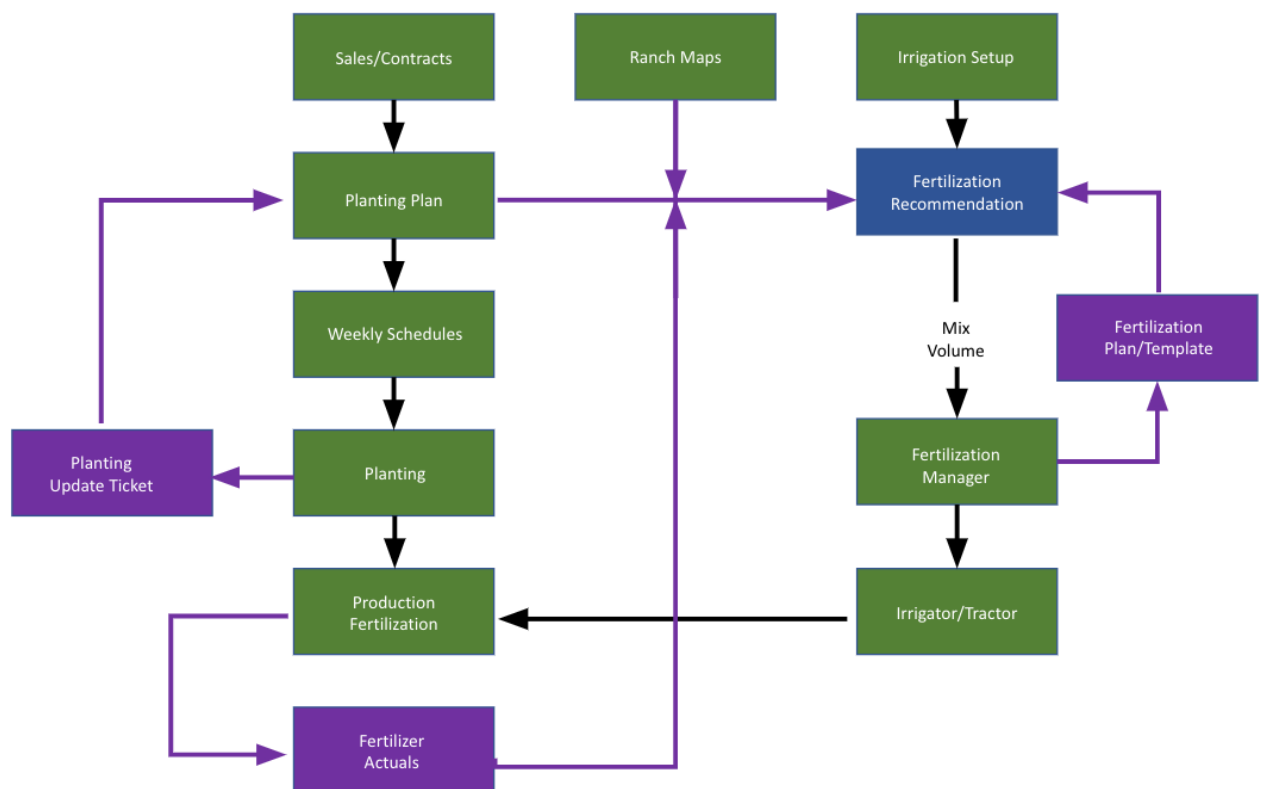


Figure 15. Fertilization process map for commercial vegetable production with additions (in purple) to incorporate precision fertilization recommendations.

2.6.3.5 API Extensions to Support Integration with Third Party Software

A specific objective of the project was to work with UCANR so they would add CropManage API features to communicate with the components of the process flow illustrated in Figures 14 and 15. To that end, we worked with the UCANR CropManage team to diagram the data flow and functionality of CropManage and how it could interface with other, third-party software capabilities. As planned, we used GeoVisual's SeedGreen web and mobile application as the test case, since GeoVisual's customers provide their planting data to SeedGreen and to other third-party data management systems.

A blocker to obtaining CropManage adoption from commercial growers has been the added effort required to enter planting data into more than one system. If they already have it in a third-party system, they will not invest the additional time to enter it a second time into CropManage. Hence a preliminary data flow and functionality diagram was developed, as shown in Figure 16, whereby data is exchanged between a third-party application in the center of the diagram. Note it is indicated as SeedGreen, but it can be any application that manages the data shown in the bullets on the left side: ranch and planting data, in-field data collection such as soil samples; actions, tasks or recommendations made for each field by management, amounts of input products applied (fertilizer, pesticides, herbicides), and any measurements of canopy cover (from UAV, satellite or in-field cameras). Much of this data may be collected by the application's mobile or web interfaces (at right), but it is all stored in the Enterprise Data warehouse - a database or other file system in the grower's facility or on a secure cloud system.

Upon looking more into the CropManage system, an additional grower concern was raised concerning data confidentiality and security. It was found that the data security practices do not meet modern, commercial software standards. This creates additional reluctance to share planting data within the system, particularly when variety-specific data is included. To work around these blockers, the CropManage team agreed to develop a "single call recommendation", whereby all grower data is maintained in its enterprise data storage, transmitted through third-party commercial applications (such as SeedGreen), and not stored in CropManage. The updated data flow diagram is shown in Figure 17. Rather than sending CropManage specific data on a grower, ranch name, lot number and crop variety, only a geographic location, crop type and irrigation date is provided, plus an estimate of canopy cover. With this information CropManage returns an irrigation recommendation.

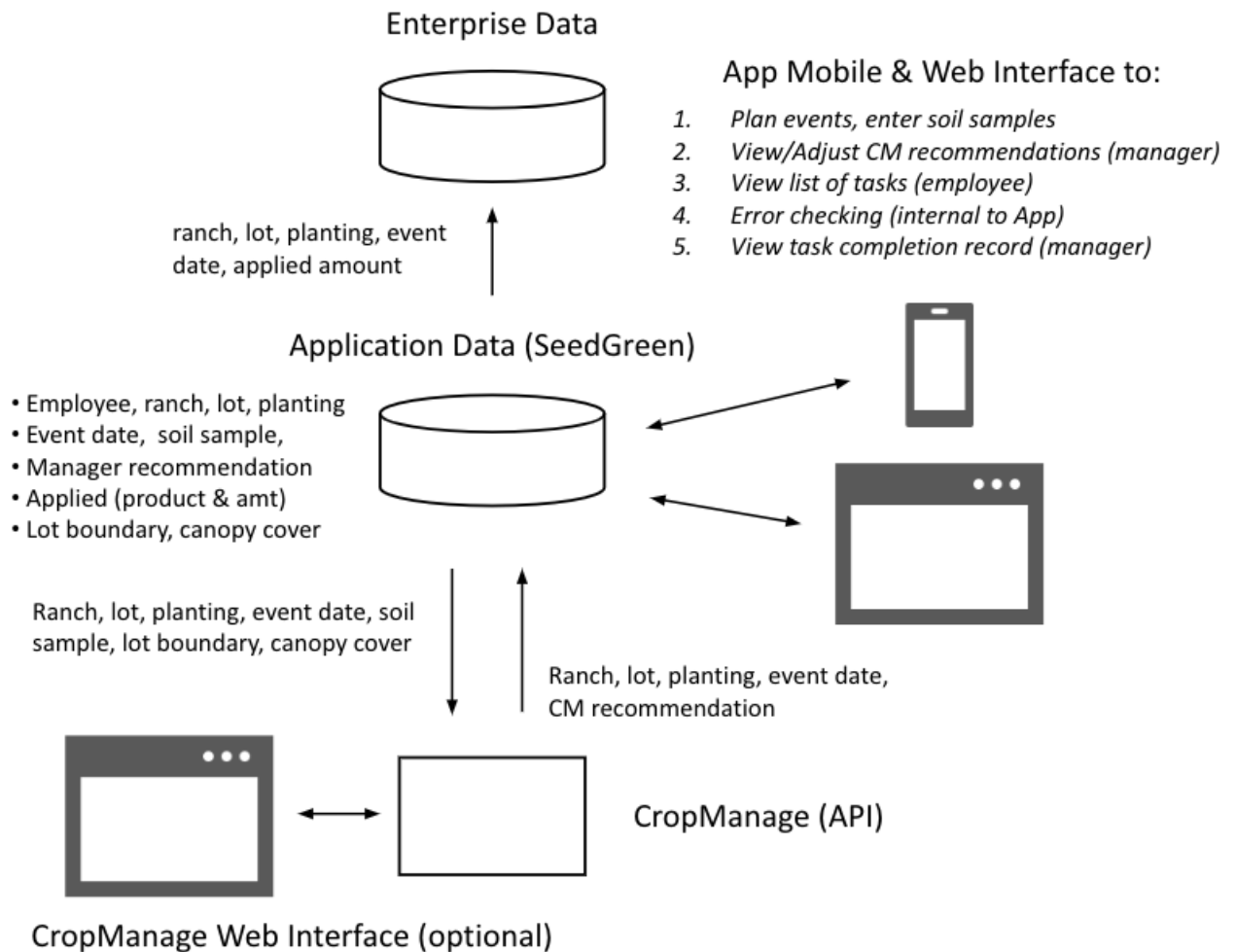


Figure 16. Preliminary data flow and functionality diagram for the CropManage irrigation and fertilization recommendation system, with data exchanged between a third-party application. The impetus for this architecture was to avoid having growers enter data about their plantings into multiple systems.

The data exchange illustrated in Figure 17 addresses the data double-entry blocker and any security concerns growers may have about storing their data in CropManage. This was the configuration the CropManage team implemented and tested for this project, and the updated, public CropManage API is available online for anyone to use.

The only remaining capability included in Figure 14 and 15 that is necessary for integrating irrigation and fertilizer recommendations into a production system is that irrigation and fertilizer “actuals” are shared back to CropManage or another recommendation system. Note that the Planting Update Ticket capability shown in Figure 14 and 15 is a “nice to have” but not critical component of an initial, basic advancement to incorporate irrigation and fertilizer recommendations into a

production system. In an optimal implementation, irrigation actuals would be obtainable through flow meter data, presumably collected with a data logger and stored in the cloud for access by the recommendation system. Short of that, in a more typical scenario given the state of the industry today, a simple irrigation timer indicating when irrigation systems are turned on and then off is a practical solution. Irrigation time information can be converted to an approximate amount of water applied to each field given the irrigation infrastructure. Optimally, irrigation crews would have an application that makes it easy to control and report on the irrigation timing. Similarly, fertilization information could be collected in the field by the relevant crew members and sent back to the fertilization recommendation system.

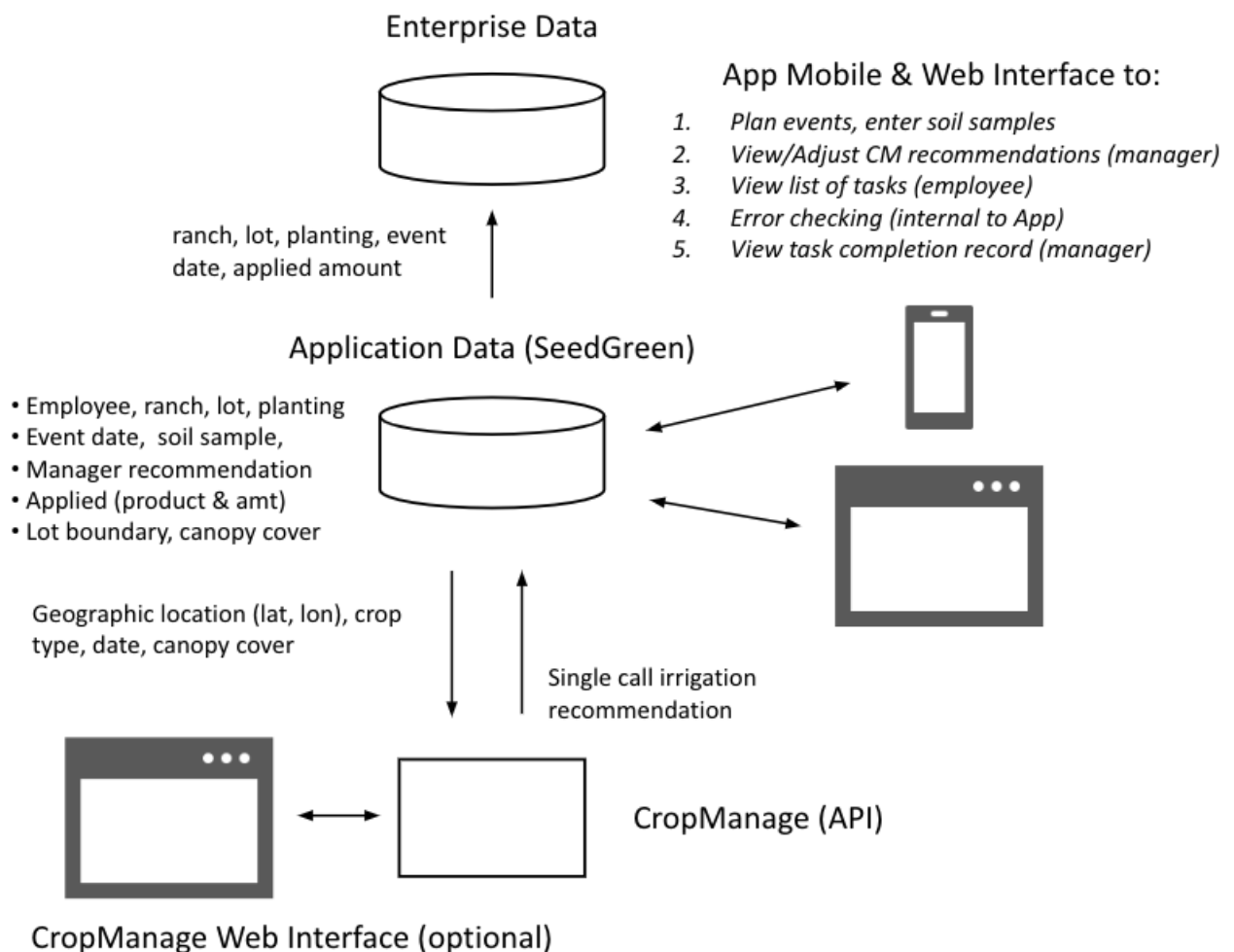


Figure 17. Final data flow and functionality diagram for the CropManage irrigation and fertilization recommendation system for this project, with data exchanged between a third-party application. In this case, planting information is not shared with CropManage. Rather, a geographic location is sent to CropManage, and not stored. An irrigation recommendation is returned.

2.6.4 Goal 4 – Promotion and Management of Stakeholder Engagement

2.6.4.1 Summary of Planned Objectives and Activities for this Goal

Planned objectives for this goal included involving at least two grower-partners in ongoing R&D during data collection in their fields and organizing a project stakeholder advisory committee to seek inputs and feedback from its members as early as possible in the project.

2.6.4.2 Summary of Actual Objectives Achieved and Activities Conducted for this Goal

During Year 1 of the project, we involved two grower-partners in California, and we organized a stakeholder advisory committee as planned. During Year 2, we expanded our grower-partner set to include additional leafy green and celery growers in the Central Coast, Central Valley and Imperial Valley regions of California and southern Arizona, the director of the Colorado Potato Administrative Council, a service provider to the Colorado potato industry, and researchers from Colorado State University and its Extension service. The geographic distribution of the grower-partner fields is shown in the maps in Figures 1 through 3 above. Throughout Year 2, we maintained periodic contact with the stakeholders, with either in-person meetings or phone conversations. In this section, we describe specifics of our grower-partners' involvement in the project, feedback received through interactions with them, and interactions we had with the advisory committee members. We have aggregated specific grower comments where necessary to maintain anonymity.

We held discussions with grower-partner ranch managers to understand their current irrigation practices and challenges. We then held semi-regular discussions with Nature's Reward's production manager, Mark Mason, during and after lettuce and celery trial studies with CropManage. We held additional periodic discussions with other grower-partners with whom we shared ongoing results from the CropManage trials.

We continued an information exchange with the University of Arizona and the Yuma Center of Excellence for Desert Agriculture (YCEDA) throughout the project. This culminated in feedback provided to YCEDA on its development of the DesertAgWISE, a mobile app management tool that is being developed to provide crop-specific recommendations that help growers fine-tune irrigation timing and quantity. It is specifically designed to provide for actual crop water needs while managing soil salinity levels for productivity and long-term sustainability. This mobile irrigation and soil salinity management app will track crop water use and soil salinity balance and allow farmers to utilize real-time satellite and weather data when making irrigation decisions based on ET and soil salinity needs.

2.6.4.3 Summary of Grower-Partner and Advisory Committee Feedback

A major challenge many growers face is that their ranches, which are several hundred acres each and contain many individual fields, typically have a single well for the entire ranch. Some ranches are fed from multiple wells supplying a reservoir. But in both cases, there is insufficient control over irrigation for individual fields at present, such as the presence of pressure regulators. It would be a large infrastructure effort (and investment) to provide flow meters for individual plots. Typically, growers plant the same crop but different varieties in contiguous fields. This makes the tracking of fertilizer, water and other inputs easier. Also, they experience less variability because each region can have a unique microclimate and cloud cover and the same soil type.

Presently, growers located in California are less concerned about amounts of irrigation water used and more attentive to fertilizer use. Current California regulations require reporting on how much fertilizer is used per crop cycle, regardless of water usage, as described in detail in the following section. Growers claim to record information on their irrigation practices, but at this point information appears to be recorded on paper in general; it is not clear what is done with the information. To determine when it's time to water a field, it depends on crop type, but typically they "kick the dirt and see if it's dry". If so, they turn the water on. They try to avoid signs of noticeable water stress on plants, as that means it's too late to avoid reductions in quality and yield.

All growers are practical and cost-conscious by necessity. A challenge with complicated irrigation systems is that they must be installed and uninstalled each time they switch crops in a field. Also, there are concerns that related technology will not be sufficiently reliable. It can be very dependent on being installed correctly. For example, if the soil becomes too dry at the tip of certain soil moisture sensors, the system requires a reset. These are impractical features in an industrial setting that must be overcome.

More than one ranch manager we interviewed sees the biggest potential for improved irrigation systems in helping reduce labor costs. As the minimum wages go up and hours get cut, predictive irrigation could help reduce the labor costs of managing irrigation. Today, monitoring every field every day is not practical. If they could monitor all ranches remotely, they could manage by exception – inspecting only those fields with a problem and requiring less irrigation labor.

A major challenge is finding and keeping labor, so they are faced with completing needed tasks as wages go up and they can depend on fewer labor hours due to

fewer available works and regulations regarding overtime pay and maximum hours worked per week. Using labor contractors adds a commission fee. In one grower's case, each one of their districts has 4 to 5 head irrigators who are highly skilled. For each of these, there are 2 to 3 lower-skilled irrigators who make minimum wage. Low-skilled irrigators typically are not well-educated, and Spanish is their first language. This impacts the practicality of technology adoption. Growers agree there will need to be a shift in hiring for the future – hiring workers with more education and a greater affinity for technology. Either they find such workers, or they invest in them through continuing education.

2.6.4.4 Environmental and Regulatory Drivers of New Irrigation Practices throughout the Western U.S.

In this section, we summarize the overarching drivers for improved irrigation practices in vegetable crop production throughout the western U.S. based on inputs received from project stakeholders and follow-up research. All stakeholders are well-aware of the water shortages in the western U.S., with overuse of Colorado River water and over drafting of aquifers exacerbated by drought conditions. Hence there are both environmental and regulatory drivers that are steadily increasing pressure on western agriculture to make smarter use of available water resources.

Regulatory pressure is shortening the mandated timelines for concrete actions, particularly in California. With California leading western agriculture, it also leads in western water consumption. According to the California Department of Water Resources, agriculture accounts for approximately 80 percent of all the water used in the state. But for agriculture producers, reducing water consumption and contamination while maintaining production levels is not a simple task. California's diverse agriculture and microclimates make it difficult to implement a one-size fits-all approach.

As is evidenced by our many conversations with producers and the experiences shared in this report, the tendency of producers to over-irrigate as "insurance" is at odds with the need for more efficient practices. Hence California's current water situation not only being impacted by the varying water needs of the diverse crops that are grown but also is constrained by past practices: the standard practices and perceptions that have been around for decades.

Tantamount to gaining producer adoption of new practices, therefore, is the need to demonstrate the cost-effectiveness of new practices - to convince irrigators and their management that investments in new irrigation infrastructure and

technology will ensure expected yields and quality while saving water and fertilizer and, optimally, making their operations more efficient and thereby profitable. While the industry generally recognizes the need for change, the existential issue is whether or not they will be able to make changes soon enough to meet imposed regulatory constraints and to ensure a sustainable future for the industry in the western U.S.

Timeframes for change in California are being defined by implementations of the state's Sustainable Groundwater Management Act (SGMA), with regional Water Quality Control Boards defined for different areas of the state as shown in Figure 18. The Central Coast region has codified their SGMA implementation in its Agriculture Order 4.0. Ag Order 4.0 was adopted early in 2021, with the regulations of most impact to vegetable growers in the region associated with the discharge of agriculture waste into the groundwater supply. As an outcome, growers will be forced to comply with a gradual ratcheting down of fertilizer application rates over time. Because this is the principal new driver for improved irrigation practices in the region, we summarize in this section the associated issues to provide context for future strategies for adoption of improved practices.

In Region 3, the Water Quality Control Board has divided implementations into three phases which will implement the requirements with different time windows depending on the criticality of water quality issues in a specific geographic area. The first groundwater phase area is assigned to areas with greater water quality impairment and higher risk to water quality compared to the other two-phase areas. Figure 19 lists the groundwater basis in Region 3 and their corresponding implementation phases. To accurately determine which phase a specific farm or ranch falls into, a grower can look up their location on the Control Board's [GeoTracker](#) website. Some farms or ranches may span over two phases, and when this does occur the phase assigned to the entity would be the earlier of the two. For example, if a farm was located in Phase 1 and Phase 2, the farms would need to adhere to Phase 1 requirements.

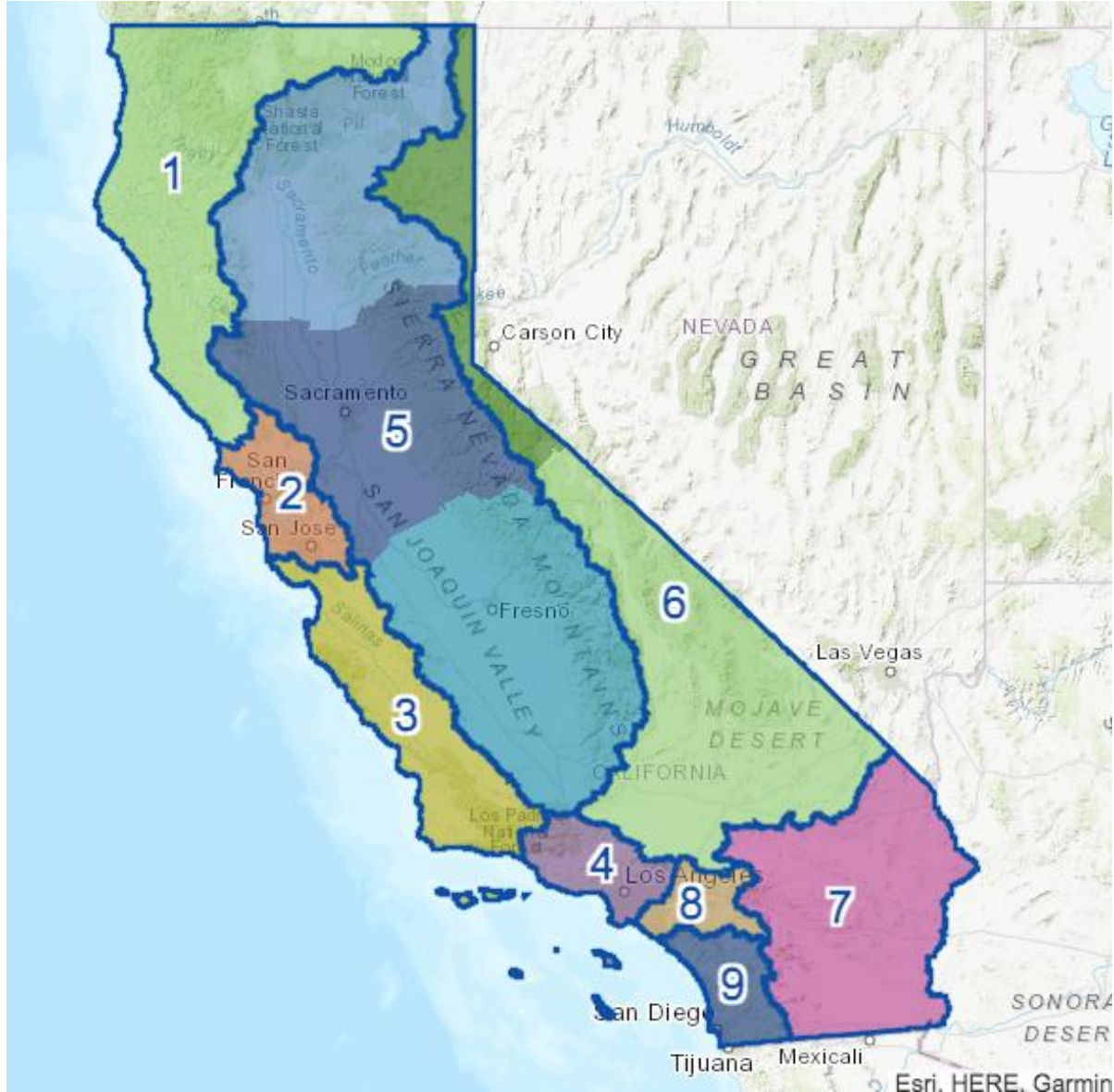


Figure 18. Geographic coverage of California's nine Regional Water Quality Control Boards. Ag Order 4.0 is the implementation of SGMA for Region 3, the Central Coast Region. For comparison, we also highlight the status of SGMA implementation for Region 5, the Central Valley Region. Together these two regions make up a large majority of the vegetable producers in California.

The two main reporting requirements for growers in Region 3 are the Total Nitrogen Applied (TNA) Report and the Irrigation and Nutrient Management Plan (INMP) report. The TNA report will require agricultural producers to collect the amount of nitrogen applied to each crop as well as estimating the amount of water that is applied to the crop. The INMP follows a stricter guideline, by requiring agricultural producers to quantify the amount of nitrogen applied, the amount of irrigation, and the amount of nitrogen that has been removed from each crop. The

stakes will be high for agricultural producers to adhere to these requirements, with violations up to \$15,000 per day of violation or \$20 per gallon of waste discharged and up to \$1,000 per day for failing or refusing to submit reports required by the Ag Order.

Groundwater Basin¹	Groundwater Phase
Gilroy-Hollister Valley - Llagas Area	Phase 1, Phase 2
Salinas Valley - Forebay Aquifer	Phase 1, Phase 2
Salinas Valley - Upper Valley Aquifer	Phase 1, Phase 2
Santa Maria River Valley - Santa Maria	Phase 1, Phase 2
Santa Ynez River Valley	Phase 1, Phase 3
Corralitos - Pajaro Valley	Phase 2
Gilroy Hollister Valley - North San Benito	Phase 2
Salinas Valley - 180/400 Foot Aquifer	Phase 2
Salinas Valley - East Side Aquifer	Phase 2
San Luis Obispo Valley	Phase 2
All Other Basins and Areas Outside of Basins	Phase 3

Figure 19. Groundwater Phase areas categorized by the groundwater basin located nearby. Phase 1 and Phase 2 areas cover the high priority areas, while Phase 3 covers the outstanding areas not included in Phase 1 and Phase 2. From the Central Coast Regional Water Quality Control Board.

By December of 2023, producers on the Central Coast will need to adhere to an overall limit of 500 pounds of TNA per year, which accounts for the applied amount of fertilizer and what has been removed through soil management practices. After hearing from producers in the area, it has become clear that many are concerned about meeting this requirement in the required timeframe. Ag Order 4.0 also requires they continue to decrease the pounds of nitrogen fertilizer applied per year and will be expected to be as low as 50 in the year 2051.

Figure 20 summarizes the reporting timelines for the 3 phases under Ag Order 4.0 and compares them to the previous requirements under Ag Order 3.0. The California Regional Water Quality Control Board approved one third party entity, Central Coast Water Quality Preservation Inc. ("Preservation Inc."), to help growers adhere to the Order's regulations. It has been advised to the public that if a grower chooses to work with Preservation Inc, the requirements will be seen as *targets* rather than *limits*. The need to better understand the Ag Order 4.0 will be vital in coming years, which is why ongoing discussions with Preservation Inc. will be important to best assist growers on the Central Coast.

Ranches	TNA¹ Monitoring Period²	TNA Report Due March 1	<u>Annual</u> INMP³ Monitoring Period²	<u>Annual</u> INMP Summary Report Due March 1
Required per Ag Order 3.0	2021 2022	2022 2023	-	-
Groundwater Phase Area 1 ⁴	-	-	Beginning 2023	Beginning 2024
Groundwater Phase Area 2	2023 2024	2024 2025	Beginning 2025	Beginning 2026
Groundwater Phase Area 3	2023 2024 2025 2026	2024 2025 2026 2027	Beginning 2027	Beginning 2028

Figure 20. Monitoring and reporting requirements for the (previous) Ag Order 3.0 and the finalized Ag Order 4.0. From the Central Coast Regional Water Quality Control Board.

The Ag Order is driving the agricultural industry in Region 3 to quickly implement ways to offset their nitrogen applications. Michael Cahn, Irrigation Advisor with the University of California Cooperative Extension, notes there are three key practices to improving fertilizer application efficiency: 1) measure residual nitrate levels in soil, 2) measure residual nitrate levels in irrigation water and adjust fertilizer application accordingly, and 3) Improve irrigation efficiency to help maintain soil nitrates at the root zone. The over-irrigation observed soon after planting and discussed in section 2.6.3.3 could be reduced with stricter attention paid to the third recommendation.

Different Regional Water Quality Control Board regions in California are rolling out SGMA implementations with different timelines, but reporting requirements are very similar. In the Central Valley, Region 5, growers are to submit Farm Evaluations and annual INMP summary reports, with an INMP worksheet also required to be completed and maintained at farming operations headquarters or primary place of business and must be provided to Board staff if requested. The Farm Evaluation will help determine what farm practices are currently being implemented and whether any improvements can be made to protect water quality. Growers in areas where groundwater is susceptible to contamination or is known to be impacted by nitrate, pesticides or other constituents associated with agriculture are required to have a certified INMP.

Practices such as installing flow meters or variable frequency pumps are key to tracking the amount of irrigation water applied to fields. The California Department of Food and Agriculture's (CDFA) State Water Efficiency and Enhancement Program (SWEET) funds the implementation of irrigation practices that reduce water and greenhouse gas emissions. The SWEET grant opportunity could help fund practices that will make adhering to the Ag Order 4.0 easier, such as purchasing a variable frequency pump drive, flow meters or soil moisture sensors. All SWEET-funded projects now must include the use of flow meters. CDFA's other grant that may be of use to help offset nitrogen is the Healthy Soils Incentives Program. This program allows California agricultural producers to implement soil management practices that improve soil health, sequester carbon, and reduce greenhouse gas emissions. Some of the practices that are eligible for this grant can be used to reduce the total nitrogen applied to the growers' field, thus helping them meet regulatory requirements.

In Arizona, researchers argue that given the efficiency of irrigation practices in southern Arizona agriculture, the limit to further improvements in efficiency are related to managing salinity levels in the soil (Sanchez, 2022). With average irrigation efficiencies, defined as the ratio of crop ET to the water applied at about 85%, this leaves a leaching fraction of 15%, which is insufficient for salt balance and results in soil salt levels building up over time. Irrigation waters used in southern Arizona come from the Colorado River, with high in situ salt levels. Excess salts in irrigation water cause osmotic effects on plants, with the plants expending more energy to draw in more brackish water. Unlike in California, where groundwater nitrate levels are of the principal concern addressed in Ag Order 4.0, nitrate levels are not as high compared to other salts, such as those containing calcium, magnesium, phosphorus, sodium and chloride. Given the increasing salt levels intrinsic to irrigation in southern Arizona, most fields are treated with a "pre-irrigation", whereby fields are watered before planting solely to leach salts below the root zone.

The Arizona Quantitative Assessment project described in Section 2.6.2.3 has been focused on addressing this issue. One element of that project is development of a mobile application to be used by growers in the region of southern Arizona. As CropManage has developed evapotranspiration and irrigation requirements models for crops in California, the irrigation-related app being developed at the University of Arizona includes models for salinity management in Arizona. As part of our collaboration with YCEDA during this FFAR project, we provided feedback to the Arizona team on their app development process. Our principal emphasis was the need for the team to focus on providing an API for integration of the app and its

underlying models with third party tools to ensure adoption in commercial agriculture.

Lastly, in Colorado, aquifer depletion is the major concern, particularly in the San Luis Valley where the bulk of the state's potatoes are grown, making Colorado the 5th largest producer of potatoes in the country. Growers in the Valley are motivated to be good water stewards, but there are limits to how precisely irrigation can be managed without sufficiently detailed data on crop conditions, soil properties and resulting soil moisture spanning the growth cycle. Irrigation scheduling techniques based on crop evapotranspiration (ET) estimates have been demonstrated to be effective at guiding potato irrigation in the Valley while ensuring expected yields. However, local variability in soil properties, terrain slope and sprinkler coverage have led to sub-optimal irrigation and crop yields. While moisture sensors could provide ground-truth data, with sensor placement based on known anomalies such as gravel channels or steep slopes, hardware costs and management complexity are a disincentive to extensive sensor use there.

Growers in the San Luis Valley have expressed interest in the use of aerial imagery, from manned or unmanned aircraft (drones), provided there is a compelling value proposition. In addition to fractional vegetation cover measurements as done for this FFAR project, automated analysis of imagery is becoming an increasingly common means to pinpoint irrigation issues such as nozzle clogs, pressure non-uniformity and water stress, covering entire fields with relatively high spatial resolution (e.g. sub-meter). Since imagery-based estimates of water stress require calibration, ideally they are combined with a sparse set of in-field moisture sensors for cost-effective monitoring. Ultimately, as variable rate center-pivot technology advances and becomes more affordable, sprinklers will be controlled based on local variability identified through a manageable number of strategically placed soil sensors and appropriately timed imagery collections, for more uniform water application.

2.6.5 Goal 5 – Stakeholder outreach and feedback events

2.6.5.1 Summary of Planned Objectives and Activities for this Goal

Planned objectives for this goal included holding a project kickoff meeting and 1 or 2 events in California and Arizona focused on stakeholder outreach and feedback.

2.6.5.2 Summary of Actual Objectives Achieved and Activities Conducted for this Goal

During Year 1 of the project, we held a virtual project kickoff meeting, a virtual outreach event in Colorado, and several meetings with project partners in Arizona to discuss priorities for outreach there. These activities are discussed in detail in the Year 1 Annual Report. As the pandemic eased during Year 2 of the project, we attended several outreach events to discuss this project, obtain grower perspectives, and formulate recommendations related to improved irrigation efficiency in the region.

CFVGA 2022 Annual Conference, February 28 – March 1, 2022:

The Colorado Fruit and Vegetable Growers Association (CFVGA) held a hybrid conference in Denver, CO. This project's PI, Jeffrey Orrey, presented a session on March 1st entitled "Remote Sensing and other Technologies for Specialty Crop Farming" focused on the role of technologies in water management and irrigation scheduling, sharing some practices from vegetable production in California and Arizona.

Tulare World Ag Expo, February 8-10, 2022:

Information was collected at the conference from over a dozen tech companies with solutions for irrigation management, and attending growers were queried for their perspectives. Principal among the technology providers was Jain Irrigation, a global irrigation systems producer and provider for agriculture. Jain and its competitor in specialty crops, Netafim, acknowledge facing many of the adoption challenges this project uncovered.

Southwest Ag Summit, February 22-24, 2022:

This annual conference is focused specifically on the particular issues associated with desert agriculture, with irrigation management a key component. As such, we met there with the team from the Arizona Quantitative Assessment project and a group of that project's stakeholders. We shared experiences from the California trials with CropManage and compared their approach regarding sharing modeling capabilities and best practices with the greater community.

Colorado State University Ag Innovation and Irrigation Stakeholders:

Throughout Year 2 of this project, we held a series of meetings with researchers and educators in CSU's College of Agriculture and CSU staff involved in their new Spur campus. Specific to Spur is a new Masters of Agribusiness and Food Innovation Management, from which there is an educational opportunity to involve young ag professionals in implementations of improved irrigation practices. There also appear to be opportunities to tie associated activities into ongoing activities

of the Irrigation Innovation Consortium, with which GeoVisual is continuing discussions on tighter collaboration. Near term plans include both involving both the Masters program students and the IIC in the Specialty Crop Block Grant focused on improved irrigation management in the San Luis Valley of Colorado.

2.7 Efforts to Ensure Scientific Rigor and Robust, Unbiased Results

FFAR feedback at the end of Year 1 of the project was that a larger data set of fields over which fractional vegetation cover samples were taken with UAV imagery and compared to satellite results was needed to ensure robust, unbiased results. To that end, we increased the data set significantly in Year 2, both in sample number and geographic distribution. Analysis of the results will be ongoing with USDA/ARS in Arizona, and the data set will be shared as a product of this project.

2.8 Key stakeholders that Could be Served by Results

UCANR and its CropManage program will be a beneficiary of this project's work in a number of ways: through the comparisons of fractional canopy cover at various spatial scales from UAV imagery to canopy cover estimates from satellites and in-field cameras; through enhancements to the CropManage platform to drive increased usage; and through increased outreach to the commercial agriculture community about CropManage and the associated capabilities available for improved irrigation and fertilization management.

The NASA SIMS program and OpenET will continue to seek validation of the satellite imagery-derived canopy fractional cover values used to estimate ET, particularly for annual vegetable crops. Mr. Forrest Melton, SIMS lead at NASA, has indicated that this work provides a foundation for integration of airborne data (from both UAV and piloted aircraft) into CropManage, which would address a key limitation of satellite-based platforms during periods when persistent fog or cloud cover create extended data gaps in the availability of satellite observations.

The University of Arizona and its Yuma Center of Excellence in Desert Agriculture (YCEDA) has been engaged for the past 3 years in a Crop ET data collection project, measuring crop water use in 13 different desert crops. As part of that project, U of A and YCEDA teams have collected extensive eddy covariance ET measurements in commercial agriculture fields in the Yuma area. Their database will benefit from the additional fractional cover measurements obtained over a large set of commercial agriculture fields that will result from this project. The U of A project has been focused on characterizing soil salinity values in the desert region and seeks to transition to an irrigation management system. Hence this project's work on making irrigation management capabilities such as CropManage

more usable by the commercial agriculture community has already begun to support that effort.

The USDA Agricultural Research Service (ARS) is engaged in related activities, particularly for a better understanding of the consumptive use water budget in the arid Western U.S. Dr. Andrew French of the USDA ARS US Arid-Land Agricultural Research Center in Maricopa, AZ will be a collaborator during Year 2 of this project. Dr. French is preparing a manuscript for publication on fractional canopy cover estimates from remote sensing and its role in accurately assessing evapotranspiration.

The Colorado potato industry will be the beneficiary of a follow-on project that is the result of this FFAR project. In collaboration with researchers at Colorado State University (CSU) and its Extension service, GeoVisual prepared a Specialty Crop Block Grant proposal that was recently awarded by the Colorado Department of Agriculture. Due to the decreasing water levels in Colorado's San Luis Valley aquifer, the proposal is to study and identify irrigation scheduling practices for growers of potatoes and leafy green crops that would result in the reduction of water use while still maintaining optimal crop yield and growth. The goal of the project is to advance irrigation scheduling capabilities in the San Luis Valley by improving timing and applied amounts of irrigation, drawing upon work done with the UCANR CropManage team and research already done at CSU. The project will utilize soil moisture sensors in order to quantify water levels, as well as using aerial imagery to observe crop canopy conditions. The data gathered from the soil moisture sensors and aerial imagery will be ingested into GeoVisual's mobile app so that growers can add crop-specific irrigation notes and record in-field observations. An objective of the project is to identify which irrigation management practices are most cost-effective and accurate. As such, the block grant is an outgrowth of the current FFAR project.

2.9 Results Dissemination to Date

The PI on this project presented a talk on the project's findings and progress at the UCANR's 2021 California Plant and Soil Conference, held virtually Feb 1-3, 2021. Over 200 attendees were logged on for the virtual presentation, including members of the UC research community and the California chapter of the American Society of Agronomy. Results also were shared during the Colorado Fruit and Vegetable Growers Association annual meeting in 2021 and the 2022 Southwest Agriculture Summit in Yuma, Arizona.

2.10 Challenges or Delays Encountered and Actions Taken

The Covid pandemic impacted the fresh produce industry and the project team's abilities to collaborate with it in a few significant but not overly detrimental ways. The principal impact was a delay in initial project start given the inaccessibility of industry partners during the initial months after the shutdown and an inability of the project team to travel to California. As such, the project kickoff meeting did not occur until June 12, 2020, and the Year 1 activities were somewhat delayed, with field data collection beginning following the kickoff in June.

The project also experienced a subsequent delay in finalizing which fields at Nature's Reward were to be used for the project. Initially, Firestone Ranch was selected, and GeoVisual was notified in mid-June about fields that had been planted the third week of May. GeoVisual therefore began flying the fields at the end of June. However, it turned out that incompatible value configurations at the initially-selected fields made it impossible for the CropManage team to instrument the fields with manifolds and flow meters, so different fields were selected for the study.

The CropManage team and Nature's Reward ultimately settled on fields for an irrigation study by early August. We concluded that the delays were due to an overall lack of coordination between the project teams, particularly since the irrigation trials were scheduled for fields that were part of Nature's Reward's regular commercial production. Had there not been travel limitations due to the pandemic, we expect more coordination would have been possible. As it was, possibly some of the delay could have been avoided by more frequent phone follow-up with Nature's Reward and the CropManage team. By Year 2, however, this project's management team involved other grower-partner stakeholders for acquisition of canopy cover imagery, which reduced the dependency on a single grower-partner.

2.11 Opportunities for Training and Professional Development

Nothing to Report.

2.12 Educational Components Involved during this Reporting Period

During Year 2, the project team established relationships with and an action plan for the future for collaboration with the CSU Masters in Agribusiness and Food Management program, from which program students will participate in follow-on work from this project. The first concrete follow-on activity will be the Specialty Crop Block Grant described above and focused on improved irrigation management activities for the fresh produce industry in the San Luis Valley of Colorado.

3 Information Products

3.1 Information Products Produced during the Project

The PI on this project presented a talk on the project's findings and progress at the UCANR's 2021 California Plant and Soil Conference, held virtually Feb 1-3, 2021. An abstract of the presentation is available on the [conference proceedings website](#), and [a recording of the session is available on YouTube](#).

The session on the Arizona Quantitative Assessment project at the 2022 Southwest Agriculture Summit and the discussion period afterward that addresses data sharing and open applications is [available on YouTube](#).

3.2 Citations for Information Products Produced During the Project

Orrey, Jeffrey, *Using UAV Data to Assist in the Production of Cool Season Vegetables*, 2021 California Plant and Soil Conference, University of California Agriculture and Nature Resources.

Brierley, Paul, French, Andrew, Sanchez, Charles and Toby Torrey, *Quantitative Assessments of Water & Salt Balance for Cropping Systems in Lower Colorado River Irrigation Districts*, 2022 Southwest Ag Summit, Yuma, AZ.

3.3 Publications or Manuscripts Accepted for Publication

None. However, Dr. Andrew French of USDA/ARS has a publication in preparation using the information product resulting from this project.

3.4 Website(s).

None.

3.5 Inventions, Patent Applications or Licenses.

None.

3.6 Confidential Information Products

None.

3.7 Additional Activities to Ensure Access to Information Products.

None.

4 Data Management

4.1 Data Generated

A dataset of 325 fractional vegetation cover measurements (fraction of a field

covered in crop vegetation versus bare soil) is available for public use. The dataset was derived from multispectral imagery collections with a UAV over iceberg lettuce fields in California and Arizona from October 1, 2020, through November 18, 2022. The fractional cover measurements have been shared with USDA/ARS. Plans are being made to make the dataset available through the OpenET website.

It is available from GeoVisual upon request: ffar-data@geovisual-analytics.com.

5 Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____Jeffrey L Orrey_____ - _____

PI Signature: __________ Date: _____01/21/2023_____

Authorized Signing Official (ASO) Name: _____Jeffrey L Orrey_____

ASO Signature: __________ Date: _____01/21/2023_____